

American Association for Bronchology and Interventional Pulmonology (AABIP) Evidence-Based Guidelines on Bronchoscopic Diagnosis and Staging of Lung Cancer

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Background: Lung cancer remains a predominant cause of cancer-related deaths worldwide, and there are notable geographic and institutional differences in both diagnostic and staging approaches. To address this, the American Association for Bronchology and Interventional Pulmonology (AABIP) convened a multidisciplinary committee to craft evidence-based and evidence-informed recommendations for diagnosing peripheral pulmonary nodules and performing convex probe endobronchial ultrasound (CP-EBUS)-guided mediastinal staging.

Methods: A modified Delphi method guided the creation and refinement of 9 Population, Intervention, Comparator, Outcome (PICO) questions. A systematic literature review, updated through March 2023, served as the basis for drafting recommendations. The panel used the Grading of Recommendations, Assessment, Development and Evaluation (GRADE) approach to assess the quality of evidence and relied on National Institute for Health and Care Excellence (NICE) language to express the strength of each recommendation. Where suitable, meta-analyses were completed; otherwise, systematic reviews and consensus among experts provided the evidence for guidance.

Results: Nine recommendations were ultimately proposed: 6 were supported by meta-analyses and 3 by systematic reviews. The topics include comparing diagnostic yield and complication rates between peripheral bronchoscopy and transthoracic needle biopsy, the use of multiple biopsy instruments and the role of rapid on-site evaluation (ROSE) during peripheral bronchoscopy, and best practices for CP-EBUS-guided mediastinal staging. Several critical considerations emerged, such as lesion size, evolving technologies in bronchoscopy, and the importance of both available resources and local expertise.

Conclusion: These guidelines aim to standardize and streamline recommendations for the bronchoscopic diagnosis and staging of lung cancer. Since rapid technological progress and observational data play significant roles in this field, ongoing research and evidence updates will be vital to refining best practices. Clinicians are advised to tailor these recommendations according to local circumstances, the unique needs of their patients, and any new findings as they develop.

Key Words: lung cancer, bronchoscopy, endobronchial ultrasound (EBUS), mediastinal staging, peripheral pulmonary nodule

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SECTION I: EXECUTIVE SUMMARY

Background and Clinical Context

Lung cancer remains the predominant cause of cancer-related mortality in the United States, with notable

geographic and institutional disparities in diagnostic yield and staging procedures. The American Association for Bronchology and Interventional Pulmonology (AABIP) has developed comprehensive guidelines addressing 2 critical domains: nonsurgical biopsy of peripheral pulmonary nodules (PPNs) and convex probe endobronchial ultrasound (CP-EBUS) for mediastinal staging.

Methodology

The guideline development process integrated a systematic literature review through March 2023, utilizing the Grading of Recommendations, Assessment, Development and Evaluation (GRADE) framework for evidence assessment and National Institute for Health and Care Excellence (NICE) methodology for recommendation formulation. A modified Delphi consensus process ($\geq 80\%$ agreement threshold) was used for recommendation validation. The multidisciplinary panel comprised pulmonologists, interventional radiologists, thoracic surgeons, and allied health professionals.

Minimum Threshold for Evidence-Based Guideline Process

The minimum criteria for producing an evidence-based recommendation and meta-analysis were at least 2 well-designed studies—randomized controlled trials (RCTs) or observational studies—that quantitatively addressed comparable interventions and outcomes, each assessed as fair or good quality.

Core Objectives

1. Standardization of diagnostic approaches for PPNs.
2. Optimization of CP-EBUS-guided mediastinal staging protocols.
3. Reduction of care variability across diverse clinical settings.
4. Mitigation of geographic and institutional outcome disparities.
5. Identification of critical research priorities.

Implementation of Framework

- Recommendations utilize NICE-guided terminology;
- “Offer” or “Use”: Interventions where empirical evidence demonstrates clear benefit-to-risk superiority, and which should be routinely offered/used in eligible patients.

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- “Consider”: Interventions where evidence suggests probable benefit for most patients, but the balance of risks/benefits or patient preferences indicates individualized decision-making.
- “Make No Recommendation”: situations where the available evidence is insufficient (either in quantity or quality) to support a definitive “use” or “do not use” statement.
- “Do Not Routinely Offer” or “Do Not Routinely Use”: Interventions where the evidence does not support routine use, or where harms may outweigh benefits for the general patient population.
- “Do Not Offer” or “Do Not Use”: Interventions where the evidence demonstrates a clear lack of benefit and harms clearly outweigh benefits.

Recommendations and Key Rationale

1. Recommendation 1: Evidence-Informed (Meta-analysis) Consider peripheral navigational bronchoscopy (PNB) with biopsy, rather than computed tomography-guided transthoracic needle biopsy (CT-TTNB), for evaluating indeterminate PPNs, taking into account patient factors, nodule characteristics, local expertise, and costs.^{1–10} Remark: Although the meta-analysis showed higher diagnostic yield with CT-TTNB based on studies of older bronchoscopic platforms, emerging data on contemporary bronchoscopic guidance techniques demonstrate comparable diagnostic yields with lower pneumothorax rates and offer procedural flexibility for concurrent lymph node staging.
2. Recommendation 2: Evidence-Based (Meta-analysis) Consider using multiple biopsy tools, rather than a single tool, during bronchoscopy to sample indeterminate PPNs.^{11–20} Remark: A meta-analysis of 10 studies demonstrates improved diagnostic yield with multiple biopsy tools without increased complications. Effect sizes remain modest, and implementation should be individualized based on patient characteristics and institutional capabilities.
3. Recommendation 3: Evidence-Based (Meta-analysis) Consider using Rapid On-Site Evaluation (ROSE) during bronchoscopy for nonsurgical biopsy of indeterminate PPNs, based on local resource availability and clinical context.^{21–26} Remark: A meta-analysis demonstrates that ROSE improves diagnostic yield without prolonging procedural time, while facilitating specimen triage for molecular testing. However, implementation requires consideration of institutional cytopathology support capabilities.
4. Recommendation 4: Evidence-Based (Meta-analysis) We make no recommendation regarding the use of diagnostic peripheral bronchoscopy versus CT-TTNB specifically to obtain tumor tissue for molecular analysis in patients with malignant PPNs.^{2,27,28} Remark: A meta-analysis of 3 comparative studies showed no significant difference between modalities for obtaining adequate tumor tissue; both methods demonstrated high success rates for molecular testing.
5. Recommendation 5: Evidence-Informed (Systematic Review) Use a systematic staging examination, rather than examining only radiographically suspicious lymph nodes, during CP-EBUS for mediastinal staging in presumed lung cancer.^{29–37} Remark: Systematic reviews demonstrate that systematic CP-EBUS staging detects 5% to 15% more occult mediastinal disease compared

with targeted sampling of radiographically abnormal nodes alone.

6. Recommendation 6: Evidence-Based (Meta-analysis) Do not routinely offer preoperative surgical mediastinoscopy to patients with a negative CP-EBUS-guided mediastinal staging examination who have risk factors for occult N2/N3 disease but no overt radiographic involvement.^{38–40} Remark: A meta-analysis indicates that post-CP-EBUS mediastinoscopy provides only marginal improvement in detecting occult N2/N3 disease without significantly reducing unexpected nodal metastases at surgery.
7. Recommendation 7: Evidence-Based (Meta-analysis) Clinicians should recognize a primary tumor size > 3 cm as a significant risk factor for occult mediastinal metastasis and consider invasive nodal staging in patients with peripheral tumors and a radiographically normal hilum and mediastinum.^{41–44} Remark: A meta-analysis demonstrates that tumors larger than 3 cm have approximately twice the likelihood of harboring occult mediastinal metastasis compared with smaller tumors. This recommendation includes risk-stratification guidance rather than a direct “use/do not use” statement.
8. Recommendation 8: Evidence Informed (Systematic Review) Do not routinely offer invasive mediastinal staging for patients with pure ground-glass nodules (GGNs) and a radiographically normal mediastinum on computed tomography (CT) and positron emission tomography (PET).^{45–53} Remark: Retrospective analyses demonstrate that pure GGNs carry an extremely low risk of mediastinal lymph node metastasis, making routine invasive staging of limited diagnostic benefit relative to procedural risks.
9. Recommendation 9: Evidence Informed (Systematic Review) We make no recommendation regarding the routine use of CP-EBUS to detect occult N1 disease before stereotactic body radiation therapy (SBRT) in patients with early-stage nonsmall cell lung cancer (NSCLC) who do not meet criteria for invasive staging.^{45,54–60} Remark: Current evidence lacks sufficient methodological rigor to demonstrate meaningful improvements in clinical outcomes, while procedural risks and potential treatment delays must be considered.

INTRODUCTION

For decades, lung cancer survival rates remained abysmal despite incremental progress in treatment. Recent advances in cancer detection and therapeutic modalities are helping to improve overall survival and reduce recurrence.⁶¹ Despite this positive trend, there are marked variations in patient outcomes between centers, regions, and socioeconomic groups.^{62–70} Much of this heterogeneity can be attributed to inconsistent adherence to best practices and guideline-concordant care.^{71,72} While variation in therapeutic approach is frequently identified as a primary determinant of disparate outcomes, appropriate use of invasive diagnostic procedures for tissue acquisition and mediastinal staging is a prerequisite for implementing optimal treatment.

The American Association for Bronchology and Interventional Pulmonology (AABIP) assembled an expert panel to develop comprehensive evidence-informed recommendations addressing 2 critical domains:

1. Nonsurgical biopsy approaches for peripheral pulmonary nodules (PPNs).

2. Convex probe endobronchial ultrasound (CP-EBUS)-guided mediastinal staging in nonsmall cell lung cancer (NSCLC).

This document synthesizes our rigorous systematic review with meta-analysis, supplemented by expert consensus where data proved insufficient for formal evidence-based guidance.

Nine distinct recommendations were formulated—6 supported by meta-analysis and 3 by systematic reviews. One meta-analysis-supported recommendation diverged from initial findings due to recent technological advances and newer data that were initially not available for inclusion into the meta-analysis. These guidelines also highlight several areas requiring further investigation, including robotic bronchoscopy, real-time intraprocedural imaging techniques, and innovative tools such as peripheral cryobiopsy.

Clinicians are urged to implement these recommendations within the context of their local expertise, patient-specific factors, and available resources. Our expectation is that these evidence-based and evidence-informed recommendations will help reduce disparities in lung cancer outcomes while improving diagnostic yield and sensitivity for PPNs and mediastinal staging assessments.

METHODOLOGY

Panel Selection

The guideline development process incorporated an executive committee from AABIP, an expert panel, and a dedicated writing group. Six pulmonologists (including 2 methodologists) comprised the executive committee that oversaw the entire guideline development process. The expert panel consisted of 9 pulmonologists, 2 interventional radiologists, and 2 thoracic surgeons.

Fourteen pulmonologists and 3 allied health professionals (2 nurse practitioners and 1 respiratory therapist) formed the writing group. Both the expert panel and writing group participated in recommendation voting, while the expert panel additionally reviewed and suggested manuscript edits for assigned Population, Intervention, Comparator, Outcome (PICO) questions.

Conflict of Interest (COI)

All nominees submitted disclosure forms before selection. Individuals holding patents or significant stock positions that might benefit from these guidelines were automatically disqualified. The executive committee managed lesser conflicts on a case-by-case basis to maintain integrity.

PICO Question Development

Initial Question Generation

Each expert panel member proposed 5 clinically relevant questions for each category:
Nonsurgical biopsy of PPNs.
Evaluation of pulmonary nodules when mediastinal staging is warranted.

Question Ranking

After reviewing for redundancy, questions were ranked 1 to 9 (1 to 3=unimportant, 4 to 6=important, 7 to 9=critically important). Questions scoring 7 to 9 received automatic selection, with additional 4 to 6 scoring questions added until reaching a total of 15.

Refining PICO Questions (Modified Delphi)

The writing group drafted preliminary PICO questions. A 2-round modified Delphi process followed with the expert panel; panelists could accept, reject, or propose revisions in round one, then voted to include or reject in round 2. Questions required 70% approval for inclusion. Nine final PICO questions emerged that targeted key aspects of diagnosing and staging pulmonary nodules.

Systematic Review and Meta-Analysis

Literature Search

Literature searches for each PICO question were conducted in 2021 with updates in March 2023 across multiple databases (including PubMed and Embase) using predefined strategies. Given the rapid pace of technological advancement in this field, a post-hoc decision was made to also consider pivotal, high-impact studies that became available after the formal literature search cutoff to ensure recommendations reflect the most current evidence. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagrams in the supplementary materials detail the screening methodology.

Screening and Selection

The writing group performed initial title screening. Two independent reviewers evaluated potentially relevant abstracts using Covidence software. Abstracts advanced to full-text review when both reviewers deemed them relevant; the guideline chair and co-chairs (RM, AC, FK, and MS) resolved disagreements. Inclusion criteria required studies to address the respective PICO elements, include ≥ 50 participants, and report specified outcomes. Two reviewers assessed full-text articles for eligibility before data extraction.

Risk of Bias and Quality Appraisal

Each included full-text article underwent bias assessment using appropriate validated tools: Quality Assessment of Diagnostic Accuracy Studies-2 (QUADAS-2) for diagnostic accuracy studies. Newcastle-Ottawa Scale for observational studies. A MeaSurement Tool to Assess Systematic Reviews-2 (AMSTAR-2) for existing systematic reviews.

Two independent reviewers assessed bias risk for each study. Disagreements were resolved through discussion or consultation with a third reviewer. Poor-quality articles were excluded. Complete risk of bias assessments appear in Supplementary Tables S1–S7, Supplemental Digital Content 1, <http://links.lww.com/LBR/A364>.

Data Extraction and Synthesis

Data extraction for each PICO question utilized standardized Excel spreadsheets capturing study design, population characteristics, interventions, comparators, outcomes, and effect measures. Due to the variable definitions of diagnostic yield, we extracted final diagnostic outcomes as determined by the individual study authors. We acknowledge that definitions of yield varied across studies (strict, intermediate, or liberal definitions) and are not necessarily consistent with the recent American Thoracic Society (ATS)/American College of Chest Physicians (ACCP) consensus definitions of diagnostic yield or accuracy.⁷³ When feasible, meta-analyses were conducted using R (version 4.4.2) and RStudio (version 2024.12.1 Build 563), employing the meta package (version 8.0-2) for both

TABLE 1. Master GRADE Summary Table for All Recommendations

Rec. #	Outcome & Comparison	# Studies	Study Design Breakdown	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Overall Certainty	Comment/Rationale
1	Diagnostic Yield: Bronchoscopy vs. TTNB	10	• RCT: 3 • Retrospective: 7	Not serious	Not serious	Serious	Serious	Suspected	Very Low	Older bronchoscopy studies show lower yields compared to CT-guided TTNB. Newer bronchoscopic methods may narrow this gap.
1	Pneumothorax (Any): Bronchoscopy vs. TTNB	10	• RCT: 3 • Retrospective: 7	Not serious	Not serious	Not serious	Not serious	None	Low	Bronchoscopy is consistently associated with significantly lower pneumothorax rates.
1	Pneumothorax Requiring Chest Tube: Bronchoscopy vs. TTNB	8	• RCT: 2 • Retrospective: 6	Not serious	Not serious	Not serious	Serious	Possible	Low	Limited events result in wide confidence intervals.
2	Diagnostic Yield: Single vs. Multiple Biopsy Tools	10	• Prospective: 4 • Retrospective: 6	Not serious	Not serious	Not serious	Not serious	Possible	Moderate	Adding an additional biopsy tool increases yield by ~10% (absolute improvement).
3	Diagnostic Yield: ROSE vs. No ROSE	6	• RCT: 3 • Prospective: 1 • Retrospective: 2	Not serious	Not serious	Not serious	Not serious	Suspected	Moderate	ROSE consistently improves diagnostic yield without significantly increasing procedure duration.
3	Procedure Time: ROSE vs. No ROSE	4	• RCT: 3 • Retrospective: 1	Not serious	Serious	Not serious	Not serious	Suspected	Low	Marked heterogeneity ($I^2 \approx 95\%$) due to variable definitions of procedure time across studies.
4	Molecular Testing Adequacy: Bronchoscopy vs. TTNB	3	• Prospective: 1 • Retrospective: 2	Not serious	Serious	Not serious	Serious	Suspected	Very Low	No clear difference in adequacy for molecular testing; small studies with heterogeneous methods.
5	Surgically Detected Occult Mediastinal Disease: Systematic vs. Targeted CP-EBUS	8	• RCT: 1 • Prospective: 3 • Retro: 4	Very serious	Serious	Not serious	Serious	Suspected	Very Low	Systematic CP-EBUS detects 5-15% more occult N2/N3 disease. Data primarily from small, retrospective studies with broad variability.
6	Missed N2/N3 Disease: CP-EBUS alone vs. CP-EBUS + Mediastinoscopy	3	• RCT: 1 • Prospective: 2	Not serious	Not serious	Not serious	Serious	Suspected	Low	Minimal additional benefit from routine mediastinoscopy after negative CP-EBUS in high-risk patients.
7	Occult Mediastinal(N2) Metastases: Tumor Size > 3 cm	4	• Retrospective: 4	Serious	Not serious	Not serious	Serious	Suspected	Very Low	Tumors > 3 cm approximately double the risk of occult N2

TABLE 1. (continued)

Rec. #	Outcome & Comparison	# Studies	Study Design Breakdown	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Overall Certainty	Comment/Rationale
8	Occult Mediastinal Disease in Subsolid Nodules: Invasive Staging vs. No Staging	9	• Retrospective: 9	Serious	Not serious	Not serious	Serious	Suspected	Very Low	involvement despite normal imaging. Pure GGOs and subsolid with CTR <0.5 rarely have lymph node involvement. Limited evidence from retrospective studies; no RCT data.
9	Occult N1 Disease Detection: CP-EBUS Before SBRT vs. No CP-EBUS	8	• Retrospective: 8	Very serious	Serious	Serious	Serious	None	Very Low	No definitive evidence that pre-SBRT CP-EBUS improves outcomes in cN0 patients. N1 rates vary (3-20%) across small retrospective studies, often involving higher-risk tumors.

Abbreviations: RCT, Randomized Controlled Trial; TTNB, Transthoracic Needle Biopsy; ROSE, Rapid Onsite Evaluation; CP-EBUS, Convex probe endobronchial ultrasound; rEBUS, Radial probe endobronchial ultrasound; GGO, Ground-Glass Opacity; SBRT, Stereotactic Body Radiotherapy

random and fixed-effects models. Results were compiled into evidence summaries.

Delphi-Based Consensus for Recommendations
Following a systematic review and quality appraisal, the writing group drafted recommendations that underwent a modified Delphi voting process involving both the expert panel and writing group. Consensus required $\geq 80\%$ agreement (Agree/Strongly Agree) with $\geq 75\%$ panel participation. Up to 3 rounds were planned if necessary. If consensus remained below 80% after 3 rounds, statements were reframed as comments (if $\geq 67\%$ agreement) or excluded ($< 67\%$). All recommendations in this guideline achieved consensus without requiring maximum rounds.

Evidence Grading and Recommendation Framework
Grading of Recommendations Assessment, Development, and Evaluation (GRADE) and National Institute for Health and Care Excellence (NICE): Basic Differences

GRADE assesses overall certainty (high, moderate, low, and very low) and helps determine strong versus conditional recommendations based on benefits, harms, values, and resources. NICE employs more direct, action-oriented wording to guide clinical practice. We applied GRADE principles for evidence certainty rating but presented final recommendations using NICE terminology.

Implementation of the NICE Framework
Recommendations utilized the following NICE-guided terms:
“Offer” or “Use”: Interventions where empirical evidence demonstrates clear benefit-to-risk superiority, and which should be routinely offered/used in eligible patients.
“Consider”: Interventions where evidence suggests probable benefit for most patients, but the balance of risks/benefits or patient preferences indicates individualized decision-making.
“Make No Recommendation”: Situations where the available evidence is insufficient (either in quantity or quality) to support a definitive “use” or “do not use” statement.
“Do Not Routinely Offer” or “Do Not Routinely Use”: Interventions where the evidence does not support routine use, or where harms may outweigh benefits for the general patient population.
“Do Not Offer” or “Do Not Use”: Interventions where the evidence demonstrates a clear lack of benefit and harms clearly outweigh benefits.

Combining GRADE evaluation with NICE language provided both robust evidence assessment and practical guidance. The certainty of evidence for each key outcome across all recommendations was assessed using the GRADE framework and is summarized in Tables 1 and 2.

Ethical Considerations
This guideline relied solely on published studies; therefore, no institutional review board approval was required. No patient-identifiable data were collected, and no new interventions were introduced. All participants adhered to ethical standards throughout the literature review process.

TABLE 2. Summary of Included Studies: Bronchoscopy Versus CT-TTNB*

References	Design and size	Interventions/tools	Diagnostic yield	Pneumothorax/chest tube	Population / Inclusion
Atkins et al (2020) ¹	Retrospective cohort; n = 135 (ENB = 135; CT-TTNB = 23)	ENB (21–22G needle) vs. CT-TTNB (17–19G coaxial)	ENB: 51.9%; CT-TTNB: 87.0%	ENB: PTX 3%; Chest tube 2%. CT-TTNB: PTX 13%; Chest tube 0%	• Single-center; retrospective • SPN ~ 28 mm average • CT-TTNA group consist of patients with – ENB followed by CT-TTNB • Excluded advanced disease or known diagnosis.
Bhatt et al (2018) ²	Retrospective cohort; n = 300 (ENB = 150; CT-TTNB = 150)	ENB (21–22G needle) vs. CT-TTNB (coaxial needle)	ENB: 66.0%; CT-TTNB: 86.0%	ENB: PTX 4%; Chest tube 2.7%. CT-TTNB: PTX 28.7%; Chest tube 1.3%	• Peripheral nodules (~ 22–24 mm) Single institution • Established ENB program
Herath et al (2023) ³	Prospective randomized; n = 36 (rEBUS-cryo = 20; CT-TTNB = 16)	rEBUS + cryobiopsy vs. CT-TTNB (coaxial needle)	rEBUS-cryo: 85%; CT-TTNB: 93.8%	rEBUS-cryo: PTX 4.2%; Chest tube 0%. CT-TTNB: PTX 44%; Chest tube 12.5%	• Peripheral pulmonary lesions ≥ 1 cm • No central lesions or bulky mediastinal lymph nodes
Hur et al (2010) ⁴	Retrospective cohort; n = 77 (CT-F-TTNB = 53; TBLB = 24)	Repeat TBLB (standard) vs. CT-F-TTNB (20–22G Chiba)	CT-F-TTNB: 90.6%; TBLB: 62.5%	TBLB: PTX 0%; Chest tube 0%. CT-F-TTNB: PTX 13.2%; Chest tube 5.7%	• Suspected malignant lesions with nondiagnostic initial bronchoscopy • Central and peripheral lesions (mean 27 ± 11 mm)
Steinfert et al (2011) ⁵	Prospective RCT; n = 48 (rEBUS-GS = 32; CT-TTNB = 16)	rEBUS + guide sheath vs. CT-TTNB (coaxial needle)	rEBUS-GS: 87.5%; CT-TTNB: 93.8%	rEBUS-GS: PTX 3.2%; Chest tube 0%. CT-TTNB: PTX 18.8%; Chest tube 0%	• Single-center RCT. • Small sample size. • Significant difference in lesion size (CT-PNB: 41 ± 21 mm vs. rEBUS-GS: 28 ± 14 mm; <i>P</i> = 0.026).
Wang et al (2016) ¹⁰	Retrospective matched; n = 108 (rEBUS-GS = 54; CT-TTNB = 54)	rEBUS-GS (20 MHz) vs. CT-TTNB (coaxial needle)	rEBUS-GS: 81.5%; CT-TTNB: 87.0%	rEBUS-GS: PTX 9.3%; Chest tube 3.7%. CT-TTNB: PTX 0%; Chest tube 0%	• Propensity-matched by lesion size/location • Mean nodule size ~ 27 mm
Wang et al (2018) ⁶	Retrospective; n = 160 (rEBUS-GS = 80; CT-TTNB = 80)	rEBUS-GS vs. CT-TTNB (standard coaxial needle)	rEBUS-GS: 65.0%; CT-TTNB: 85.0%	rEBUS-GS: PTX 1.3%; Chest tube 0%. CT-TTNB: PTX 17.5%; Chest tube 1.3%	• Peripheral nodules < 3 cm, suspicious for malignancy • No bulky lymph nodes or advanced disease.
Zhang et al (2017) ⁷	Retrospective; n = 522 (rEBUS-GS = 262; CT-TTNB = 260)	rEBUS-GS vs. CT-TTNB (standard coaxial needle)	rEBUS-GS: 59.6%; CT-TTNB: 84.6%	rEBUS-GS: PTX 0.4%; Chest tube 0%. CT-TTNB: PTX 11.5%; Chest tube 4.3%	• Single institution • Lesion size 10–50 mm, peripheral • No advanced disease
Zhu et al (2019) ⁹	Retrospective; n = 529 (rEBUS-GS = 250; CT-TTNB = 279)	rEBUS-GS vs. CT-TTNB (standard coaxial needle)	rEBUS-GS: 78.2%; CT-TTNB: 94.6%	rEBUS-GS: PTX 0.4%; Chest tube NR. CT-TTNB: PTX 34.5%; Chest tube NR	• Large single-center cohort • rEBUS primarily for lesions > 80 mm from chest wall; CT-TTNB if ≤ 80 mm
Zhang et al (2023) ⁸	Retrospective; n = 114 (ENB = 21; CT-TTNB = 93)	ENB vs. CT-TTNB (standard coaxial needle)	ENB: 76.1%; CT-TTNB: 73.1%	ENB: PTX 0%; Chest tube 0%. CT-TTNB: PTX 16.1%; Chest tube 6.5%	• Nodules in “middle third” of lung • Lesion size 8–40 mm
Lentz et al (2025) ^{74†}	Prospective multicenter RCT; n = 234 (ENB = 121; CT-TTNB = 113)	ENB with digital tomosynthesis vs. CT-TTNB (coaxial needle)	ENB: 79.0%; CT-TTNB: 73.6%	ENB: PTX 3.3%; Chest tube 0.8%. CT-TTNB: PTX 28.3%; Chest tube 8.0%	• Single center; small ENB sample • Peripheral nodules (1–3 cm) • Multicenter; randomized design • Key modern ENB trial

*CT-guided procedure terminology varies in source studies; cited papers may refer to it as CT-TTNA (Aspiration); CT-PNB (Percutaneous Needle Biopsy); CT-NAB (Percutaneous Needle Aspiration) CT-TTB (Transthoracic Biopsy); CT-guided lung puncture; or CT-guided biopsy/sampling.

†Study published after the March 2023 literature search cutoff; included by post-hoc panel consensus due to high relevance and impact. Study not included in the main meta-analysis.

CT-F-TTNB indicates computed tomography with fluoroscopy-guided transthoracic needle biopsy; CT-TTNB, computed tomography-guided transthoracic needle biopsy; ENB, electromagnetic navigational bronchoscopy; GS, guide sheath; NR, not reported; PTX, pneumothorax; rEBUS, radial endobronchial ultrasound; RCT, randomized controlled trial; SPN, solitary pulmonary nodule; TBLB, transbronchial lung biopsy.

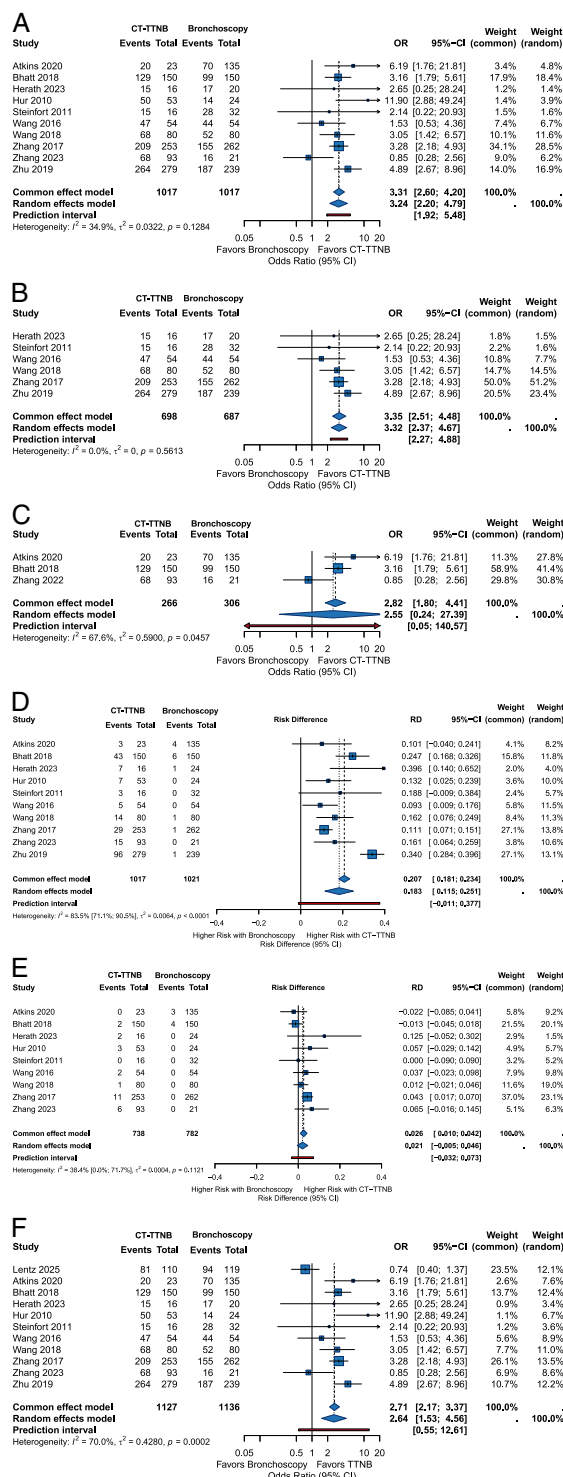


FIGURE 1. (A) (Recommendation 1). Diagnostic Yield (All Modalities) – CT-TTNB vs. Bronchoscopy. Forest plot showing the odds ratios (ORs) with 95% CIs comparing the diagnostic yield of CT-guided transthoracic needle biopsy (CT-TTNB) vs. bronchoscopy across 10 studies. The squares represent individual study estimates, with the size reflecting each study's weight under the common-effect model. The horizontal lines through the squares indicate the 95% CI. The diamonds at the bottom represent the pooled effect estimate using a random-effects

model. Heterogeneity was moderate ($I^2 = 34.9\%$). Overall, CT-TTNB demonstrated a significantly higher diagnostic yield compared with bronchoscopy [random-effects OR: 3.24 (2.20–4.79), $P < 0.0001$]. (B) (Recommendation 1). Diagnostic yield (Radial EBUS-Only) – CT-TTNB vs. Bronchoscopy. Forest plot displaying ORs and 95% CIs for diagnostic yield of CT-TTNB versus bronchoscopy in 6 studies using radial endobronchial ultrasound (rEBUS). The squares indicate study-specific estimates, sized according to the weight under the common-effect model; whiskers show the 95% CI. The diamond indicates the pooled effect under a random-effects model. Heterogeneity was minimal ($I^2 = 0.0\%$). Overall, CT-TTNB had a higher diagnostic yield than rEBUS bronchoscopy alone [random-effects OR: 3.32 (2.37–4.66), $P = 0.0003$]. (C) (Recommendation 1). Diagnostic yield (ENB-Only) – CT-TTNB vs. Bronchoscopy. Forest plot comparing diagnostic yield of CT-TTNB vs. electromagnetic navigation bronchoscopy (ENB) alone across 3 studies. Each square indicates an individual study's odds ratio (OR) with 95% CI. The diamond represents the random-effects pooled estimate. Heterogeneity was moderate ($I^2 = 67.6\%$). Under the random-effects model, the pooled OR was 2.55 (0.24–27.39), $P = 0.2328$, reflecting high variance and no significant difference in yield. (D) (Recommendation 1). Pneumothorax Incidence – CT-TTNB vs. Bronchoscopy. Forest plot showing risk differences (RD) for pneumothorax incidence in CT-TTNB vs. bronchoscopy across 10 studies. Squares show study-level RDs with 95% CIs, and the diamond indicates the random-effects pooled estimate. There was substantial heterogeneity ($I^2 = 83.5\%$). Overall, CT-TTNB has a higher pneumothorax rate than bronchoscopy [RD: 0.18 (0.12–0.25), $P = 0.0002$]. (E) (Recommendation 1). Pneumothorax requiring chest tube – CT-TTNB vs. Bronchoscopy. Forest plot of the risk difference (RD) for pneumothorax requiring chest tube placement in patients undergoing CT-TTNB versus bronchoscopy, synthesized from 9 studies. The squares depict individual RDs with 95% CIs, and the diamond represents the random-effects pooled estimate. Heterogeneity was moderate ($I^2 = 38.4\%$). The random-effects estimate shows a modest, nonsignificant increase in chest tube placement with CT-TTNB [RD: 0.02 (–0.00 to 0.05), $P = 0.096$]. (F) (Recommendation 1). Diagnostic yield (Including VERITAS) – CT-TTNB vs. Bronchoscopy. Forest plot comparing diagnostic yield for CT-TTNB vs. bronchoscopy in 11 studies, including the VERITAS trial. Individual study ORs with 95% CIs are shown as squares with whiskers, and the random-effects pooled estimate is shown as a diamond. Substantial heterogeneity was present ($I^2 = 70.0\%$). CT-TTNB had a significantly higher diagnostic yield [random-effects OR: 2.64 (1.53–4.56), $P = 0.0002$].

SECTION IV: RECOMMENDATIONS

Recommendation 1: Evidence-Informed (Meta-Analysis)

PICO Question

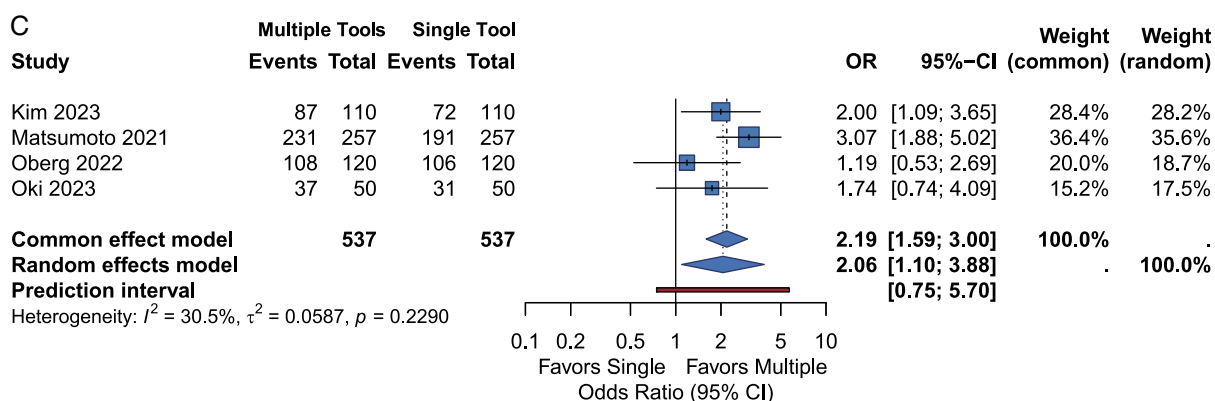
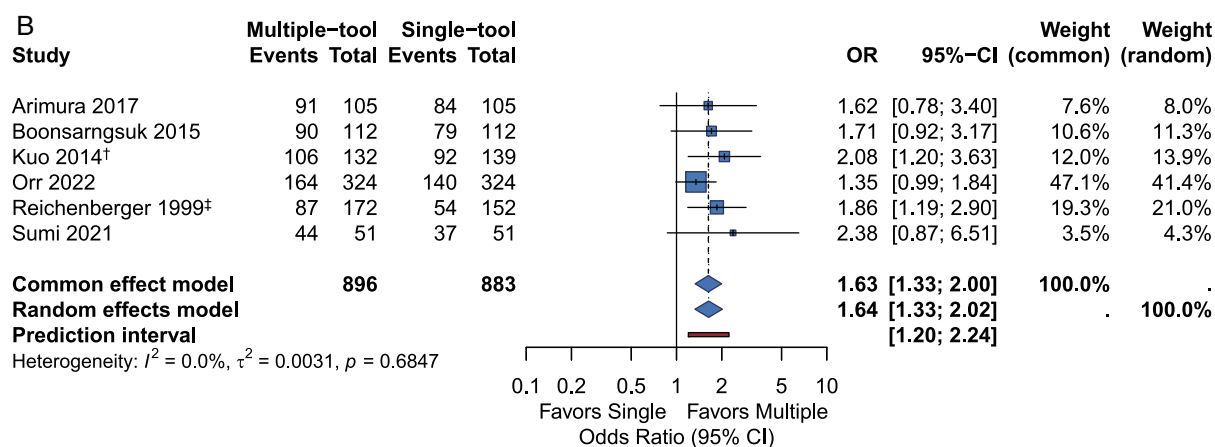
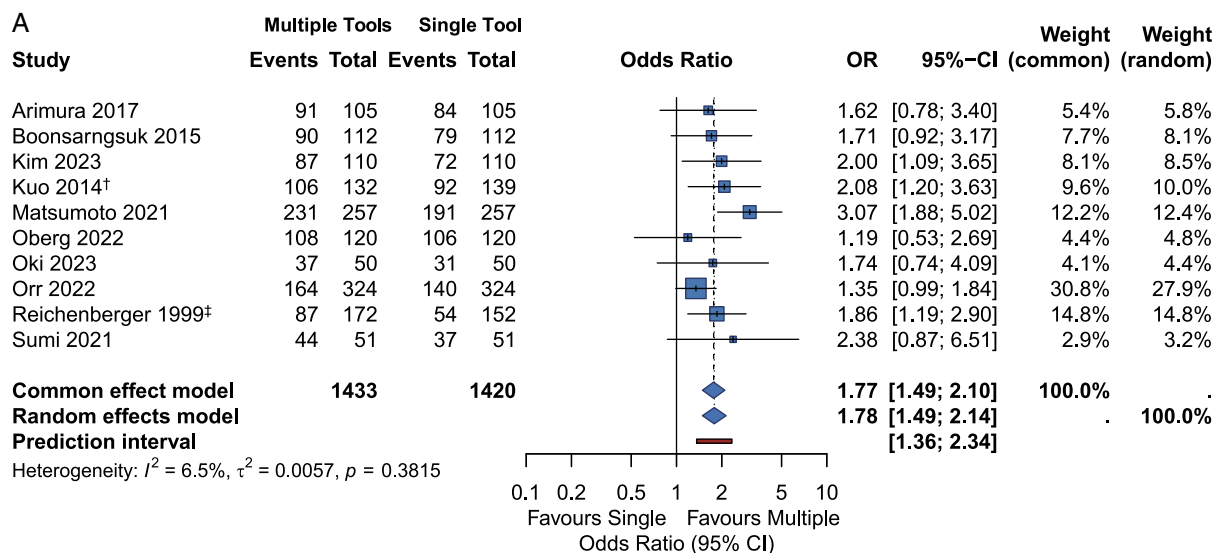
In patients with indeterminate PPNs, should peripheral navigational bronchoscopy (PNB) or computed tomography (CT)-guided transthoracic needle biopsy (CT-TTNB) be used for tissue diagnosis?

Recommendation

Consider PNB with biopsy, rather than CT-TTNB, for evaluating indeterminate PPNs, taking into account patient factors, nodule characteristics, local expertise, and costs.

Background

PPNs can represent either benign or malignant entities, and accurately characterizing the lesion is essential for appropriate and timely management.⁷⁵ CT-TTNB has



traditionally been the gold standard for PPN biopsy. While some advancements aimed at reducing complications (eg, use of gelatin powder or hydrogel plugs for tract embolization) have been introduced, there has been little progression in the core biopsy technique or significant improvement in diagnostic yield over time.⁷⁶ In contrast, guidance

modalities used for PNB have evolved significantly. Radial-probe endobronchial ultrasound (rEBUS) with 2-dimensional (2D) fluoroscopy, electromagnetic navigation bronchoscopy (ENB), and, more recently, robotic-assisted platforms have progressively allowed for more precise targeting and accurate diagnosis of PPNs during

FIGURE 2. A (Recommendation 2). Diagnostic yield: single vs. multiple tools (overall). Forest plot of odds ratios (ORs) and 95% CIs comparing the diagnostic yield of a single-tool approach vs. a multiple-tool (combined) approach, based on 10 studies ($n = 1572$). Squares illustrate individual study ORs, with size proportional to study weight. The diamond represents the random-effects estimate. Heterogeneity was low ($I^2 = 6.5\%$). Both pooled estimates indicated higher diagnostic yield with multiple tools [random-effects OR: 1.78 (1.49–2.14), $P < 0.0001$]. “single tool” arm represents the diagnostic yield reported for the single highest-performing tool in each study, compared against the overall diagnostic yield achieved using all tools combined in the “multiple tools” arm. † Single-tool yield (TBBW: 92/139) and multitool yield (TBBW + Brushing: 103/132) were derived from 2 separate patient cohorts, explaining the different denominators. ‡ Single-tool yield (TBNA: 54/152) reflects only the subset of 172 total lesions where TBNA was performed and assessed (attempted in 153). Multitool yield (full work-up: 87/172) includes all lesions. The denominator mismatch arises because TBNA was not attempted in all cases. B (Recommendation 2). Diagnostic yield (conventional tools subset): multiple vs. single tools. Forest plot of 6 studies evaluating single vs. multiple conventional biopsy tools. The squares reflect each study’s OR and 95% CI; the diamond represents the random-effects pooled estimate. Heterogeneity was low ($I^2 = 0.0\%$). The combined approach demonstrated superior diagnostic yield [random-effects OR: 1.64 (1.33–2.02), $P = 0.0018$]. “Single tool” arm represents the diagnostic yield reported for the single highest-performing tool in each study, compared against the overall diagnostic yield achieved using all tools combined in the “multiple tools” arm. † Single-tool yield (TBBW: 92/139) and multitool yield (TBBW + Brushing: 103/132) were derived from 2 separate patient cohorts, explaining the different denominators. ‡ Single-tool yield (TBNA: 54/152) reflects only the subset of 172 total lesions where TBNA was performed and assessed (attempted in 153). Multitool yield (full work-up: 87/172) includes all lesions. The denominator mismatch arises because TBNA was not attempted in all cases. C (Recommendation 2). Cryobiopsy subset: multiple vs. single tools. Forest plot comparing single-tool to multiple-tool approaches specifically using cryobiopsy in 4 studies. Squares show study-level ORs with 95% CIs. The diamond represents the random-effects estimate. There was minimal heterogeneity ($I^2 = 30.5\%$). Overall, using multiple tools with cryobiopsy yielded a higher diagnostic success [random-effects OR: 2.06 (1.10–3.88), $P = 0.0356$]. “Single tool” arm represents the diagnostic yield reported for the single highest-performing tool in each study, compared against the overall diagnostic yield achieved using all tools combined in the “multiple tools” arm.

bronchoscopy. Incorporation of real-time imaging techniques such as digital tomosynthesis (DT) and cone-beam computed tomography (CBCT) has further enhanced procedural capabilities.⁷⁷

Summary of Evidence

Ten comparative studies ($n = 2043$) were included in the meta-analysis—7 retrospective and 3 prospective, including 2 randomized controlled trials (RCTs) (Supplementary Fig. S1, Supplemental Digital Content 1, <http://links.lww.com/LBR/A364>).^{3,5,6} All transthoracic interventions utilized CT guidance. Peripheral bronchoscopic procedures used rEBUS guidance in 6 studies and electromagnetic navigation in 3.^{1,2,8} Risk of bias varied, reflecting differences in study design and patient selection criteria.

- (1) Diagnostic yield: The pooled yield for CT-TTNB procedures reached 87.0%, versus 67.0% for PNB. In our analysis, CT-TTNB had higher odds of diagnostic success [OR: 3.24 (2.20–4.79), $P < 0.0001$; Fig. 1A].^{1–10} Two RCTs, both utilizing rEBUS guidance, reported equivalent diagnostic rates compared with CT-TTNB (Fig. 1B). However, these RCTs were small and enrolled patients with relatively large lesions, averaging over 3 cm.^{3,5} In 5 studies that evaluated nodules ≤ 2 cm, CT-TTNB maintained a proportional diagnostic yield advantage of $\sim 20\%$ over PNB.^{2,4,6–8} Fig. 1C summarizes ENB studies, and relevant details of all studies appear in Table 2.
- (2) Complications: Pneumothorax was the most common complication, occurring significantly more frequently with CT-TTNB procedures than with PNB, with a pooled risk difference of 0.18 (0.12–0.25), $P = 0.0002$ (Fig. 1D). Nine studies further stratified pneumothorax rates based on chest tube requirements. While the trend again favored PNB, the effect did not reach statistical significance [RD: 0.02 (–0.00 to 0.05), $P = 0.096$; Fig. 1E].^{1,2,4–8,10}
- (3) Recent advances: A recently completed multicenter RCT (VERITAS) compared ENB with DT to CT-TTNB. This study demonstrated a 79% overall diagnostic yield with ENB compared with 73.6% for CT-TTNB.⁷⁴ However, CT-TTNB had a significantly higher risk of complications, including pneumothorax

requiring intervention (11.5% vs. 0.8% with PNB). When this trial was incorporated into the meta-analysis, CT-TTNB maintained a higher overall yield (85.7% vs. 68.4% for PNB), though the gap narrowed relative to earlier data [OR: 2.64 (1.53–4.56), $P = 0.0002$; Fig. 1F].

Early data on robotic-assisted bronchoscopy (RAB) suggest improved diagnostic yields and reduced pneumothorax rates compared with earlier PNB modalities. A key difference between the VERITAS study and previous ENB studies was the addition of real-time intraprocedural imaging with DT. As technologies like CBCT become further integrated into PNB practice, the diagnostic yield gap between bronchoscopic and transthoracic biopsy modalities may further narrow. The additional cost and radiation exposure of these techniques warrant careful consideration.

Rationale

Although older data show CT-TTNB provides a higher diagnostic yield, newer bronchoscopic modalities and applications may help achieve comparable yields while maintaining a lower risk of pneumothorax. Bronchoscopy also provides the flexibility of simultaneous mediastinal staging with CP-EBUS and multinodule sampling in a single procedure. The panel’s recommendation drew on a meta-analysis of existing literature largely reflecting older PNB technologies. However, the factors detailed above justify a more nuanced recommendation, prompting an “Evidence-Informed” stance.

Implementation Considerations

- Clinical efficacy: Newer PNB modalities with real-time imaging may narrow the diagnostic gap with lower complications; additional data are needed.
- Multitarget capabilities: Bronchoscopy enables sampling of multiple lesions in one session (including additional lung nodules and CP-EBUS-guided lymph node staging).
- Combining procedures: PNB can be paired with CP-EBUS and surgical resection under a single anesthesia event, improving efficiency of lung cancer management.⁷⁸
- Cost and resource utilization: Consider institutional resources before large equipment and staffing investments.⁷⁹

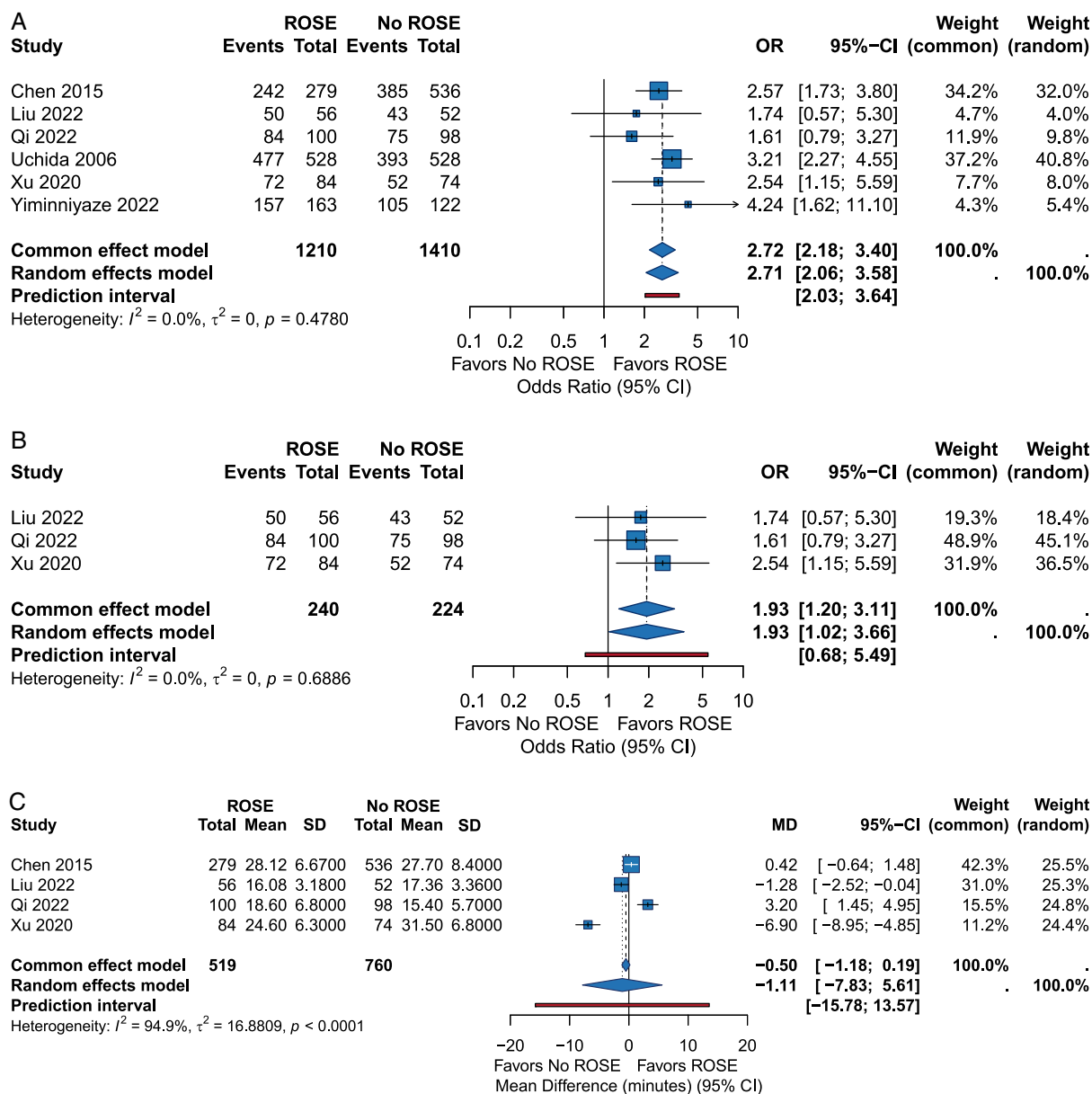


FIGURE 3. (A) (Recommendation 3). Diagnostic yield: ROSE vs. No ROSE (All Studies). Forest plot comparing rapid on-site cytologic evaluation (ROSE) vs. no ROSE in 6 studies ($n = 2710$) for diagnostic yield. Individual ORs and 95% CIs are shown as squares with whiskers; the diamond is the random-effects pooled estimate. Heterogeneity was negligible ($I^2 = 0.0\%$). Overall, ROSE significantly improved diagnostic yield [random-effects OR: 2.71 (2.06–3.58), $P = 0.0002$]. (B) (Recommendation 3). Diagnostic yield: ROSE vs. No ROSE (RCTs Only). Forest plot of 3 randomized controlled trials (RCTs) comparing ROSE with no ROSE for diagnostic yield. Squares display study-level ORs with 95% CIs; the diamond indicates the random-effects pooled estimate. Heterogeneity was negligible ($I^2 = 0.0\%$). The random-effects model suggests a significant benefit for ROSE [OR: 1.93 (1.02–3.66), $P = 0.0475$]. (C) (Recommendation 3). Procedure duration (ROSE vs. No ROSE). Forest plot comparing procedure duration (mean difference, MD) in minutes for ROSE vs. no ROSE in 4 studies. Each square shows the study-specific MD with 95% CI; the diamond represents the random-effects pooled estimate. Substantial heterogeneity was observed ($I^2 = 94.9\%$). Neither analysis showed a statistically significant difference in procedure time [random-effects MD: −1.11 (−7.83 to 5.61), $P = 0.6368$].

TABLE 3. Summary of Included Studies: Single Versus Multiple Biopsy Tools for Peripheral Pulmonary Lesions

References	Study design and sample size	Interventions/tools	Diagnostic yield (single best vs. all combined)	Complications/adverse events	Key population / inclusion
Arimura et al (2017) ¹¹	Prospective; n = 130 suspected lung CA	TBB + brush + TBNA	TBB 80% All 86.7%	Not described	• Mean lesion ~30 mm • Adequate for molecular testing
Boonsarngsuk et al (2015) ¹²	Prospective; n = 112 PPLs	TBB + brush smear + cell block + rinse + BAL	TBB 70.5% All: 80.4%	Pneumonia (1); no severe complications	• Mean lesion ~23.5 mm • BAL also used for infection screening
Kim et al (2023) ¹³	Retrospective; n = 110 PPNs ≤ 30 mm	TBB, Brush, TBNA (PeriView FLEX) + Cryobiopsy (1.1 mm)	Cryo 65.5% All: 79.1%	PTX (1), infection (1), bleeding (9)	• Mean lesion ~22 mm • ~8.6% had moderate bleeding requiring tamponade
Kuo et al (2014) ¹⁴	Retrospective; n = 271 PPLs	TBB + Wash ± Brush	TBB 66.2% All: 80.3%	PTX (7), significant bleeding (22)	• Mean lesion ~35 mm • Combined brushing/wash → +10% yield
Matsumoto et al (2021) ¹⁵	Retrospective; n = 257 PPLs	TBB + TBNA + Brush + Cryobiopsy (1.9 mm)	Cryo 74.3% All 89.9%	Severe bleeding (3), PTX (2), abscess (2)	• Mean lesion ~23 mm • ~9% incremental yield from adding cryobiopsy
Oberg et al (2022) ¹⁶	Retrospective; n = 112 (robotic-assisted)	TBNA + TBB + Cryobiopsy (1.1 mm)	Cryo 88.3% All 90%	PTX (6), chest tube (3), bleeding (3)	• Mean lesion 22 mm • High specimen adequacy with cryobiopsy
Oki et al (2023) ¹⁷	Prospective; n = 50 PPNs ≤ 30 mm	TBB + Cryobiopsy (1.1 mm)	Cryo 62% All 74%	Minor bleeding in 94% (self-limited), 1 admission	• Mean lesion ~18 mm • Manageable cryo-related bleeding
Orr et al (2022) ¹⁸	Prospective; n = 324 PPLs	GenCut Core Needle + TBB	TBB 43.2% All 50.6%	PTX (5), bleeding (2)	• Tissue adequate for advanced testing
Reichenberger et al (1999) ¹⁹	Retrospective; n = 172 PPLs	TBNA + TBB + wash + brush	TBNA 35.5% All: 50.6%	PTX (1), self-limited bleeds (12)	• TBNA significantly helps yield in certain sizes/locations
Sumi et al (2021) ²⁰	Retrospective; n = 51 PPNs ≤ 30 mm	Ultrathin + TBNA (PeriView FLEX) + TBB	TBB 72.5% All 86.3%	PTX (2), no chest tube	• Mean lesion ~23 mm • TBNA + TBB synergy for “Adjacent” lesions

BAL indicates bronchoalveolar lavage; Cryo, cryobiopsy; PPL, peripheral pulmonary lesion; PPN, peripheral pulmonary nodule; PTX, pneumothorax; TBB, transbronchial biopsy; TBNA, transbronchial needle aspiration.

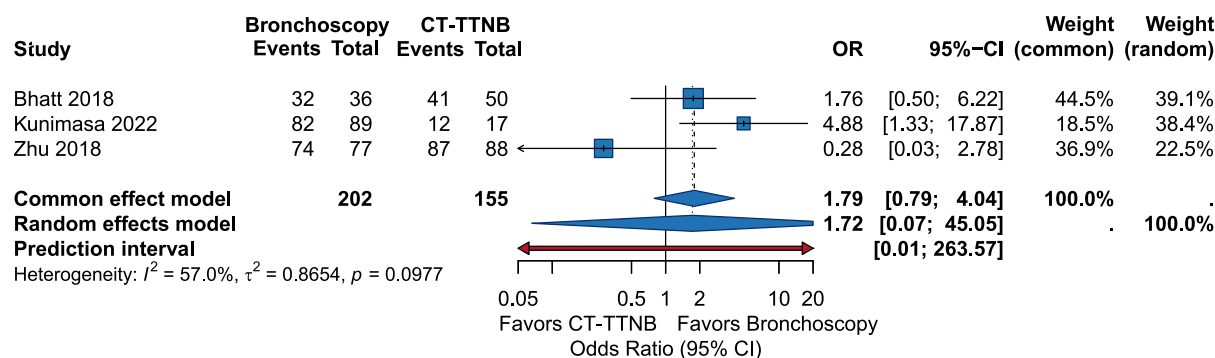


FIGURE 4. (Recommendation 4). Molecular testing adequacy – bronchoscopy vs. CT-TTNB. Forest plot of 3 studies ($n = 357$) comparing adequacy for molecular testing in bronchoscopic samples vs. CT-TTNB specimens. Squares illustrate individual ORs and 95% CIs; the diamond represents the random-effects pooled estimate. Moderate heterogeneity ($I^2 = 57.0\%$). Neither model reached statistical significance favoring one approach over the other [random-effects OR: 1.72 (0.07-45.05), $P = 0.55$].

Recommendation 2: Evidence-Based (Meta-Analysis)

PICO Question

During bronchoscopic evaluation of indeterminate PPNs, should multiple biopsy tools, rather than a single biopsy tool, be used when sampling the lesion?

Recommendation

Consider using multiple biopsy tools, rather than a single tool, during bronchoscopy to sample indeterminate PPNs.

Background

Bronchoscopic sampling of PPNs has evolved since the introduction of flexible bronchoscopy in the 1970s.⁸⁰ Traditional tools like transbronchial biopsy (TBB), transbronchial needle aspiration (TBNA), brushing, and bronchoalveolar lavage (BAL) offer complementary advantages.^{81,82} Forceps biopsy may yield lower diagnostic rates in nodules without a “bronchus sign,”²⁰ while needle-based approaches can sample airway-adjacent lesions more effectively. Cryobiopsy provides larger, better-preserved tissue samples.¹³

Summary of Evidence

A total of 10 studies ($n = 1572$) comparing single-tool with multitool PPN sampling were included for pooled analysis (Fig. S2, Supplemental Digital Content 1, <http://links.lww.com/LBR/A364>). Overall, adding a second or third tool increased diagnostic yield from ~63% to ~73%, an absolute improvement of about 10% [OR: 1.78 (1.49-2.14), $P < 0.0001$; Fig. 2A]. Table 3 provides further details of the included studies.¹¹⁻²⁰

In a subgroup analysis of 6 studies using only traditional tools (forceps, needle, and brush), multitool strategies again outperformed single-tool approaches [OR: 1.64 (1.33-2.02), $P = 0.0018$; Fig. 2B].^{11,12,14,18-20} TBB was the highest-yield tool in 5 of the studies. Despite evidence elsewhere,^{82,83} only 1 of 4 studies using TBNA found it to be the top performer.¹⁹

Cryobiopsy was utilized in 4 studies, and it was the highest-yield single tool for all 4 (range: 65% to 88%). Adding another tool increased yield by 2% to 14%. This was confirmed by a subgroup analysis (Fig. 2C), which favored using multiple tools over cryobiopsy alone [OR: 2.06 (1.10-3.88), $P = 0.036$].^{13,15-17}

Rationale

Despite heterogeneity across studies, the additional cost and time required for multiple tools are likely outweighed by the need to prevent repeat biopsy procedures after a nondiagnostic result. Lesion location appears to influence tool efficacy, with TBNA and cryobiopsy performing better for lesions adjacent to the airway compared with TBB or brushing. Registry data (AQuIRE) show that TBB exclusively diagnosed ~11% of cases, with TBNA adding an additional 6%.⁸³ As demonstrated in the NAVIGATE trial, omitting either method from a multitool approach can significantly reduce true-positive rates.⁸⁴

Implementation Considerations

- **Diagnostic yield:** A multitool approach should increase diagnostic yield by ~10% compared with a single tool, likely reducing costs and complications associated with repeat procedures.
- **Performance comparison:** Cryoprobe and TBNA may outperform TBB and brushing when lesions are adjacent to rather than within the airway. The primary tool’s baseline yield influences the magnitude of the benefit (eg, 2% to 17% gain).
- **Individualized approach:** A universal approach is likely unnecessary; tool choice should be individualized based on lesion location, size, onsite cytologic analysis, molecular marker needs, and operator experience.

Recommendation 3: Evidence-Based (Meta-Analysis)

PICO Question

Should rapid on-site evaluation (ROSE) be used during bronchoscopy for the evaluation of indeterminate PPNs?

Recommendation

Consider using ROSE during bronchoscopy for non-surgical biopsy of indeterminate PPNs, based on local resource availability and clinical context.

Background

Prior investigations have established that ROSE improves both diagnostic yield and tissue optimization for ancillary testing

TABLE 4. Rapid On-Site Evaluation (ROSE) Versus No ROSE in PPNs										
References	Study design	Population	Biopsy method	Pts		Dx yield w/ ROSE	Dx yield w/o ROSE	Complications w/ ROSE	Complications w/o ROSE	Time w/ ROSE (min)
				w/ ROSE	w/o ROSE					
Chen et al (2015) ²¹	Retrospective	815 PPNs	rEBUS TBB /Brush	279	536	86.7% (Total)	71.8% (Total)	Not reported	Not reported	28.1 ± 6.7
Liu et al (2022) ²²	Prospective RCT	108 PPNs	CT-guided Perc Core Needle	56	52	89.3% (Total)	82.7% (Total)	13.5% PTX, 21.2% bleed	13.5% PTX, 21.2% bleed	16.1 ± 3.2
Qi et al (2022) ²³	Prospective RCT	198 PPNs	rEBUS + VBN TBB	100	98	84% (Total)	75.5% (Total)	0% overall	0% overall	18.6 ± 6.8
Uchida et al (2006) ^{24*}	Prospective obs.	657 PPNs	Fluoro-guided Bronch	657	657	90.3% (Malign)	74.4% (Malign)	Not reported	Not reported	Not reported
Xu et al (2020) ²⁵	Prospective RCT	158 PPNs	rEBUS TBB	84	74	85.7% (Total)	70.3% (Total)	27% mild bleed	27% mild bleed	24.6 ± 6.3
Yiminnyaze et al (2022) ²⁶	Retrospective	285 suspected LC	CT-guided TT Core Needle	163	122	96.3% (Total)	86.1% (Total)	13.1% PTX, 9.0% bleed	13.1% PTX, 9.0% bleed	Not specified

*Uchida et al 2006²⁴ used a single cohort. The 74.4% figure is the initial yield before ROSE feedback, the 90.3% figure is the yield after ROSE within the same 528 malignant cases; no separate control arm or time data were provided. Bronch indicates bronchoscopy; CT, computed tomography; Dx, diagnostic; Fluoro, fluoroscopy; LC, lung cancer; Malign, malignant; min, minutes; Obs., observational; Perc, percutaneous; PPN(s), peripheral pulmonary nodule(s); PTX, pneumothorax; RCT, randomized controlled trial; rEBUS, radial-probe endobronchial ultrasound; ROSE, rapid on-site evaluation; TBB, transbronchial biopsy; TT, transthoracic; VBN, virtual bronchoscopic navigation

during mediastinal lymph node sampling with CP-EBUS.^{85,86} However, its precise utility for peripheral pulmonary nodules (PPNs) remains less defined, owing to concerns regarding resource utilization, procedural cost, and potential extension of procedure time by on-site cytopathological review.^{87–90} Moreover, it is uncertain how direct tool-in-lesion confirmation provided by contemporary localization technologies such as CBCT might impact the incremental value of ROSE.

Summary of Evidence

Six studies were analyzed (n=2710)—3 RCTs, 2 retrospective studies, and 1 prospective observational investigation (Supplementary Fig. S3, Supplemental Digital Content 1, <http://links.lww.com/LBR/A364>).^{21–26} Meta-analysis demonstrated that ROSE significantly enhanced diagnostic yield compared with procedures without ROSE [OR: 2.71 (2.06–3.58), P=0.0002; Fig. 3A]. Examination of the RCT subset revealed a significant benefit for ROSE [OR: 1.93 (1.02–3.66), P=0.048; Fig. 3B]. Procedure duration showed negligible overall difference under random-effects analysis (Fig. 3C), with a mean difference of –1.11 minutes (–7.83 to 5.61), P=0.637 and high heterogeneity. Complication occurrence, including pneumothorax and mild hemoptysis, varied across studies. A summary of these studies is provided in Table 4.

Rationale

ROSE appears to enhance diagnostic yield without significantly extending procedural time. Nevertheless, several important contextual factors warrant consideration. All analyzed studies were conducted in Asian centers and amid increasing adoption of real-time imaging for “tool-in-lesion” confirmation. While ROSE offers immediate assessment of specimen adequacy, advanced imaging modalities may provide similar confidence of lesion access, potentially limiting ROSE’s incremental benefit. Only 2 studies specifically utilized CT-TTNB, leaving this question unresolved. Institutions with robust cytopathology services may derive particular value from ROSE implementation, especially for high-risk diagnostic cases.

Implementation Considerations

- Staffing and workflow integration: ROSE requires a trained on-site cytopathologist (or tele-ROSE support) for timely slide preparation and interpretation.^{91,92}
- Resource allocation analysis: Initial implementation costs require evaluation against potential reductions in repeat procedures.
- Expertise assessment: Centers lacking sufficient cytopathology personnel might explore tele-ROSE solutions or emerging artificial intelligence (AI)-driven cytology platforms.⁹³
- Research priorities: Further investigation is warranted regarding the interaction between ROSE and intra-procedural advanced lesion localization technologies.

Recommendation 4: Evidence-Based (Meta-Analysis)

PICO Question

In patients with malignant PPNs, should bronchoscopy or CT-TTNB be used for obtaining tumor tissue for molecular analysis?

TABLE 5. Bronchoscopy Versus TTNB for Molecular Analysis in Malignant PPNS

References	Study design and comparison	Platform and handling	Genes interrogated	Dx yield for molecular analysis	Major complications	Key findings
Bhatt et al (2018) ²	Retrospective; n = 86 (50 CT-TTNB, 36 ENB)	Sanger and allele-specific PCR; FISH (ALK); ISH (HER2)	EGFR; KRAS; BRAF; HER2 amplification; ALK fusion	CT-TTNB: 41/50 (82.0%) ENB: 32/36 (88.9%)	CT-TTNB: PTX 28.7%; bleed 16.7% ENB: PTX 4.0; bleed 3.3%	All 5 targets successfully assayed; ENB showed higher adequacy despite smaller samples
Kunimasa et al (2022) ²⁷	Retrospective; n = 116 (17 CT-TTNB, 99 TBB)	Oncomine Comprehensive Assay v3 (DNA + RNA NGS; 161 genes); AllPrep extraction; Qubit QC; ROSE + H&E for bronchoscopy	161-gene DNA/RNA panel (SNVs, CNVs, fusions)	CT-TTNB: 12/17 (70.6%) TBB: 82/99 (82.8%)	Not reported	Driver mutations in 54% overall (EGFR 24%, KRAS 11%, MET 3.5%); tissue-confirmation workflow improved adequacy
Zhu et al (2018) ²⁸	Prospective; n = 155 (88 CT TTNA, 77 rEBUS)	RT-PCR/ARMS; break-apart FISH	<i>EGFR</i> ex 18-21; <i>ALK</i> ; <i>ROS1</i>	CT-TTNA: 87/88 (98.9%) rEBUS-GS: 74/77 (96.1%)	CT-TTNA: PTX 37.9; hemorrhage 20.9% rEBUS-GS: PTX 0.7; hemorrhage 0.7%	<i>EGFR</i> mutations in 31 cases; <i>ALK</i> / <i>ROS1</i> ≤ 2%

ARMS indicates amplification refractory mutation system; CNV, copy-number variation; CT-TTNA, CT-guided Transthoracic Needle Aspiration; CT-TTNB, CT-guided transthoracic needle biopsy; ENB, electromagnetic navigation bronchoscopy; FISH, Fluorescence In Situ Hybridization; rEBUS-GS, guide-sheath radial endobronchial ultrasound; H&E, hematoxylin & eosin (stain); ISH, in situ hybridization; NGS, next-generation sequencing; NSCLC, nonsmall cell lung cancer; PTX, pneumothorax; ROSE, rapid on-site evaluation; RT-PCR, reverse-transcriptase polymerase chain reaction; rEBUS, radial endobronchial ultrasound; SNV, single-nucleotide variant; TBB, transbronchial biopsy.

TABLE 6. Levels of Systematic Staging for CP-EBUS-Guided Mediastinal Assessment

Level of thoroughness	Definition	Key criteria/features	Rationale
Targeted sampling	Biopsy only nodal stations that appear morphologically abnormal (CT ≥ 10 mm) or FDG avid.	• Sample every clearly abnormal node. • No biopsies from morphologically normal / PET-negative stations	Simplifies staging & reduces procedure time but misses occult N2/N3 disease in up to 20% of cases. ^{34,103}
Minimum standard (systematic)	Sequential sampling of the core mediastinal chain in descending order N3 → N2 → N1 whenever a node is ≥ 5 mm or PET-positive.	• Sample 4R, 4L, 7 (± 2R/2L if feasible) when PET-positive or ≥ 5 mm short-axis on EBUS. • ≥ 3 passes or ROSE per nodal station*. • Cease deeper sampling after a malignant higher-level node (optional).	Balances sensitivity and efficiency, recommended by multiple guidelines. ^{102–104}
Thorough exam	As per Level 2, plus a systematic evaluation of every accessible mediastinal and hilar station (i.e., biopsy all nodes ≥ 5 mm and < 5 mm if sonographically suspicious) in the same N3→N2→N1 sequence.	• Biopsy all nodes ≥ 5 mm plus < 5 mm if any suspicious sonographic traits†. • Include hilar 10/11 when accessible.	Increases sensitivity for occult N2/N3 disease at the cost of longer procedure time. ^{101,105–107}
Complete systematic staging	As per Level 3, plus posterior/inferior nodes accessed by EUS-B/EUS after the CP-EBUS sequence (adds stations 8 and 9; optional 5/6).	• All Level 3 steps, plus: – Switch to esophageal route after CP-EBUS to sample stations 8 & 9 (± 5/6).	Provides the highest sensitivity, detecting ~5–15% additional occult disease versus CP-EBUS alone, but requires dual modality skills and added time. ^{30,104,108}

*With ROSE: the onsite cytologist may reduce unnecessary extra passes only after confirming both a definite diagnosis and enough cellularity for downstream testing (eg, molecular profiling). In practice that still means at least 3—and frequently 4 or more passes per nodal station.¹⁰²

†Suspicious sonographic features for nodes < 5 mm include: (1) round morphology (short to long axis ≈ 1); (2) sharply demarcated margin; (3) heterogeneous or mottled echotexture; (4) coagulation necrosis sign (avascular hypoechoic focus); (5) loss of central hilar structure; and/or (6) abnormal Doppler vascularity—grade II/III chaotic flow or bronchial artery inflow—adapted from Nakajima et al.¹⁰⁹

CP-EBUS, convex-probe endobronchial ultrasound; EUS, endoscopic ultrasound; EUS-B, esophageal ultrasound performed with an EBUS scope; CT, computed tomography; FDG, fluorodeoxyglucose; N1 / N2 / N3, nodal Levels; PET, positron emission tomography; ROSE, rapid on-site cytologic evaluation.

TABLE 7. Systematic Versus Targeted CP-EBUS

Study	Study design	Sample size	Key population	Tools/ interventions compared	Systematic definition	Targeted definition	Diagnostic yield/efficacy	Complications/ adverse events	Additional key outcomes
Block et al (2010) ²⁹	Retrospective cohort	93	NSCLC needing mediastinal staging	CP-EBUS # LN stations tested	LN ≥ 5 mm from N3→N1	Only FDG-avid LN on PET/CT	Systematic sampling detected all mediastinal disease once ≥ 4 LN stations were biopsied (60% found at 1st station; 90% by the third)	Not specifically reported	Feasibility of multistation sampling
Cordovilla et al (2021) ³⁰	Retrospective cohort	45	iN0–N1 NSCLC	CP-EBUS ± EUS	LN ≥ 5 mm from N3→N1 plus EUS if negative	CP-EBUS alone, suspicious LN only	95% sensitivity vs. 80% CP-EBUS alone; EUS found extra 6.6% missed by CP-EBUS	No severe complications	Combining CP-EBUS +EUS beneficial in PET–CT-normal mediastinum
Crombag et al (2019) ³¹	Prospective multicenter	229	Suspected resectable NSCLC	CP-EBUS+EUS-B vs. PET/CT-targeted CP-EBUS	LN ≥ 8 mm in stations 2R,2L,4R,4L,7	Sample only PET/CT abnormal nodes	Sensitivity: 73% targeted → 77% systematic (+4%) → 82% EBUS + EUS B (+9%); clinically relevant extra info in 10% of patients	Not specifically reported	Single scope systematic CP-EBUS + EUS B detects one extra clinically important nodal finding per 10 patients
Evison et al (2016) ³²	Prospective multicenter	642	NSCLC, pre/post ESTS quality standard	CP-EBUS with or w/o formal LN sampling	Minimum of 3 LN stations (4R,4L,7) targeted	Only LN suspicious on imaging	Sensitivity ~85–86%; only 16% met ≥ 3 station sampling; no overall sensitivity improvement	Not reported	Standardization reduced variability, but not absolute sensitivity
Sanz-Santos et al (2018) ³⁴	Retrospective	107	NSCLC with PET-suspected N2	Systematic CP-EBUS vs. hypothetical targeted	LN ≥ 5 mm from contralateral N3→N1	LN if FDG-avid on PET–CT	94% sensitivity, 96% accuracy, discovered 13% additional N3/N2b vs. targeted	1.8% complication rate	Systematic approach changed management by revealing more advanced disease
Serra et al (2020) ³⁵	Retrospective	174	cN0–N1 NSCLC	Systematic CP-EBUS	LN ≥ 5 mm from contralateral (N3) to N1	Not applicable	12% unsuspected N2; 0% N3; sensitivity 84%, specificity 100%, NPV 96.7%	No major complications	High negative predictive value for cN0–N1
Steinfort et al (2024) ^{36*}	Prospective multicenter	155	Locally advanced (cIII) NSCLC for RT	Systematic CP-EBUS ± EUS-B vs. PET-based staging	LN ≥ 6 mm from N3→N1 systematically	Only FDG-avid LN	~12% had PET-occult LN; CP-EBUS changed RT field in ~10%	No serious CP-EBUS-related events	Endosonography more accurate than PET alone for RT planning
Sullivan et al (2022) ³⁷	Prospective RCT	38	cN0–N1 NSCLC needing CP-EBUS staging	Targeted (omit triple-normal LN) vs. systematic	LN stations 4R, 4L, 7 systematically	Skip LN if normal by CT, PET, CP-EBUS	Missed disease 5.45% (targeted) vs. 0% (systematic); procedure time 3 vs. 19 min	No adverse events reported	Feasibility data for omitting biopsy in “triple-normal” LN

*Study published after the March 2023 literature search cutoff; included by post-hoc panel consensus due to high relevance and impact.

CP-EBUS indicates convex probe endobronchial ultrasound; EUS, endoscopic ultrasound; EUS-B, endoscopic ultrasound performed with an EBUS scope; LN, lymph node; FDG, fluorodeoxyglucose; iN0: Imaging node-negative; cN0: Clinically node-negative; PET-CT, positron emission tomography–computed tomography; NSCLC, nonsmall cell lung cancer; N2/N3, advanced nodal disease; ESTS, European Society of Thoracic Surgeons; NPV, negative predictive value; RCT, randomized controlled trial; RT, radiotherapy.

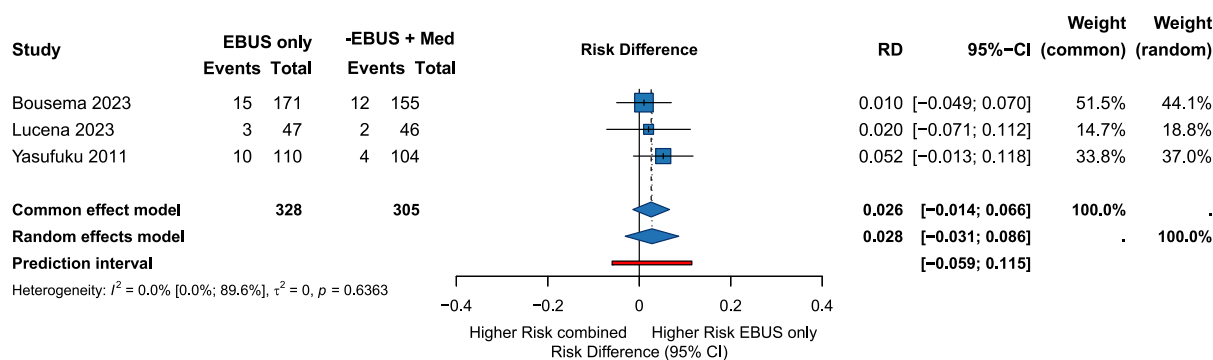


FIGURE 5. (Recommendation 6). Missed N2/N3 Disease – EBUS-only vs. combined approach. Forest plot assessing the risk difference (RD) in missed N2/N3 nodal disease when using CP-EBUS-only staging versus a combined approach, across 3 studies (EBUS only $n = 328$; combined $n = 305$; total across arms $n = 633$; 562 unique patients). Squares denote study-level RDs with 95% CIs. The diamond represents the random-effects pooled estimate. Heterogeneity was minimal ($I^2 = 0.0\%$). There was no statistically significant difference in missed N2/N3 disease between CP-EBUS alone and a combined strategy [RD: 0.03 (–0.03 to 0.09), $P = 0.18$].

Recommendation

We make no recommendation regarding the use of PNB versus CT-TTNB specifically to obtain tumor tissue for molecular analysis in patients with malignant PPNS.

Background

Efficient and comprehensive tumor characterization at diagnosis is critical for maximizing the benefits of targeted and immunotherapies. As personalized treatments advance into earlier-stage disease, minimally invasive diagnostic methods must reliably provide adequate samples for genetic and molecular testing. Insufficient tissue can delay or compromise optimal management.

Summary of Evidence

A meta-analysis of 3 studies ($n = 357$, Fig. S4, Supplemental Digital Content 1, <http://links.lww.com/LBR/A364>) comparing PNB with CT-TTNB for providing adequate tumor tissue for molecular analysis showed no significant difference between the approaches [OR: 1.72 (0.07–45.05), $P = 0.55$, Fig. 4].^{2,27,28} The overall evidence certainty was considered very low due to heterogeneous methods, small sample sizes, and nonrandomized designs (Table 5). All 3 studies reported comparable success rates, with bronchoscopy showing a small nonsignificant advantage in ^{22,27} and CT-TTNB in the third.²⁸ Tissue adequacy for molecular testing exceeded 90% in both groups.

Rationale

On the basis of this limited evidence, PNB and CT-TTNB appear to have high and comparable tissue adequacy rates for molecular studies, though these results apply to cases with positive biopsies. (For detailed discussion of diagnostic efficacy, see Recommendation 1.) While CT-TTNB might yield larger core samples for molecular testing, tissue adequacy depends on multiple factors, such as tumor size, location, necrotic proportion, and procedural aspects, including needle choice, number of passes, and efficiency of immunohistochemical staining.^{94–96} This suggests the specific method (bronchoscopic or transthoracic) may be less important than optimizing lesion selection, procedural protocols, and operator expertise.^{94,97–100} Randomized trials are needed to clarify which approach

provides the best balance of diagnostic yield, tissue adequacy for molecular marker testing, safety, and cost-effectiveness.

Implementation Considerations

- Choice of sampling approach: Limited evidence suggests no clear advantage for either modality; choice should be guided by the likelihood of obtaining adequate diagnostic material with minimal patient risk.
- Concurrent mediastinal evaluation: Bronchoscopy may be preferable when mediastinal staging (CP-EBUS) is required in the same procedure.
- Advanced molecular panels: Both approaches can achieve advanced molecular diagnostics in experienced hands. Careful attention to patient selection and procedural details (including multiple passes and meticulous tissue handling) is crucial.
- Future directions: Additional high-quality, prospective data are needed to define the optimal tissue acquisition strategy for molecular testing.

Recommendation 5: Evidence-Informed (Systematic Review)

PICO Question

Does systematic CP-EBUS evaluation improve important clinical outcomes compared with targeted CP-EBUS evaluation in patients with presumed lung cancer?

Recommendation

Use a systematic staging examination, rather than examining only radiographically suspicious lymph nodes, during CP-EBUS for mediastinal staging in presumed lung cancer.

Background

Studies reveal significant variability in the thoroughness of CP-EBUS-directed mediastinal staging across centers and operators.^{34,101,102} This variation can lead to missed occult metastases in normal-sized or nonfluorodeoxyglucose (FDG)-avid nodes, potentially affecting treatment decisions and outcomes.^{29,32} Guidance is needed about the extent of CP-EBUS evaluation required to

TABLE 8. Confirmatory Mediastinoscopy Versus No Mediastinoscopy After Negative CP-EBUS Staging

Study	Study design and sample size	Key population/inclusion criteria	Tools/interventions compared	Diagnostic yield/treatment efficacy	N2/N3 missed by CP-EBUS alone	N2/N3 missed by CP-EBUS +Med	NNT for additional N2/N3 w/Med	Complications/AEs
Bousema et al (2023) ³⁸	RCT; n = 360	Resectable NSCLC w/ high LN risk, negative CP-EBUS/EUS	Immediate resection vs. confirmatory mediastinoscopy post-negative CP-EBUS/EUS	Unforeseen N2: 8.8% immediate vs. 7.7% mediastinoscopy. CP-EBUS alone noninferior.	8.8% (15/171)	7.7% (12/155)	91	Major morbidity: 12.9% (immediate) vs 15.4% (Med)
Lucena et al (2023) ³⁹	Prospective; n = 49	NSCLC with negative mediastinum on PET-CT but risk factors	Integral Mediastinal Staging: CP-EBUS → mediastinoscopy if negative	~6.4% missed LN with CP-EBUS alone vs. 4.3% with full IMS.	6.4% (3/47)	4.3% (2/46)	49.1	None
Yasufuku et al (2011) ⁴⁰	Prospective trial; n = 153	Suspected NSCLC; all had invasive staging before surgery	CP-EBUS-TBNA vs. Mediastinoscopy, then final surgical path	~9.1% occult N2/N3 after negative CP-EBUS vs. 3.8% after mediastinoscopy.	9.1% (10/110)	3.8% (4/104)	19.1	No EBUS-related events; (2.6%) minor Med complications

AEs indicate adverse events; CP-EBUS, convex probe endobronchial ultrasound; EUS, endoscopic ultrasound; IMS, integral mediastinal staging; LN, lymph node; Med, mediastinoscopy; NNT, number needed to treat; NPV, negative predictive value; NSCLC, nonsmall cell lung cancer; PET-CT, positron emission tomography-computed tomography; RCT, randomized controlled trial; TBNA, transbronchial needle aspiration.

optimize mediastinal staging in the setting of possible lung cancer.

Levels of Thoroughness for Mediastinal Staging

CP-EBUS [and when applicable, endoscopic ultrasound (EUS)] sampling for staging purposes can be categorized into 4 levels of breadth: targeted sampling, minimum standard (Systematic), thorough exam, and complete systematic staging. These range from sampling only abnormal or highly suspicious lymph nodes to sampling any lymph node in every accessible station through both CP-EBUS and EUS (Table 6).^{30,34,102–109}

While debate persists about whether more exhaustive staging justifies the added procedural complexity, consensus exists that at least the “Minimum Standard” (ie, systematic) approach should be performed in most cases. More extensive methods might be reserved for higher-risk scenarios.^{30,110}

Summary of Evidence

A systematic review identified 5 articles for qualitative synthesis, but criteria for pooled analysis were not met (Fig. S5, Supplemental Digital Content 1, <http://links.lww.com/LBR/A364>). Consequently, a broader review discovered 8 studies of variable methodology that demonstrated consistent findings: systematic CP-EBUS approaches detect 5% to 15% more occult N2/N3 disease compared with targeted sampling.^{29,31,34} Even among patients with a radiographically normal mediastinum (cN0–N1), occult nodal metastases can occur in up to 24% of cases.^{30,35} Although sampling multiple nodal stations may slightly prolong procedures, it helps avoid missing disease that could alter treatment plans.^{29,32,36} One small feasibility RCT suggested that deferring biopsy of “triple-normal” lymph nodes (normal size, no PET uptake, negative ultrasound features) might be safe, but larger trials are needed to confirm this approach.³⁷ Table 7 provides more details on relevant studies.

Rationale

While evidence supporting systematic CP-EBUS staging comes largely from retrospective and observational studies, findings consistently suggest increased detection of nodal metastasis with systematic sampling compared with targeting only suspicious nodes. Patients with radiographically normal mediastina may still harbor occult disease, and earlier identification can significantly influence treatment decisions. Though systematic staging demands greater technical skill and adds procedural time, it improves overall staging accuracy, alters treatment plans, and decreases the risk of discovering unexpected N2/N3 disease during surgery.

Implementation Considerations

- Standardization requirements: Systematic CP-EBUS necessitates comprehensive mediastinal anatomic knowledge and adherence to standardized protocols.^{32,34}
- Competency considerations: Systematic mediastinal staging requires more advanced training than targeted sampling. Not all CP-EBUS operators need full competency in systematic staging but should clearly document the sampling approach used.
- Resource implications: Sampling additional stations increases procedural time, sedation requirements, and overall costs. High-volume centers typically demonstrate

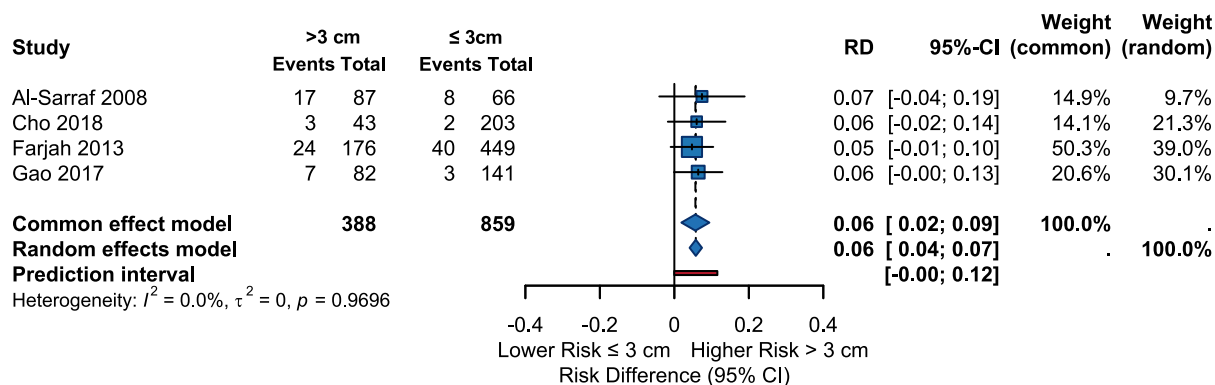


FIGURE 6. (Recommendation 7). Lesion size ≤ 3 cm vs. > 3 cm. Forest plot of risk differences (RDs) in a dichotomized lesion size comparison (> 3 cm vs. ≤ 3 cm) across 4 studies ($n = 1247$). Squares denote individual study estimates with 95% CIs; the diamond represents the random-effects pooled estimate. No heterogeneity was detected ($I^2 = 0.0\%$). Overall, the larger lesion size group (> 3 cm) had a higher event rate [RD: 0.06 (0.04-0.07), $P = 0.0016$].

better adherence to systematic sampling and improved staging performance.³⁰

- Patient selection criteria: Systematic CP-EBUS is generally recommended for patients with abnormal lymph nodes on imaging, central tumors, or tumors larger than 3 cm, given a higher risk for nodal metastases.^{110,111} Systematic staging may be omitted in select cases, such as small peripheral T1N0 lesions without CT/PET abnormalities or when there is obvious mediastinal infiltration (in which specific nodal sampling is less relevant).¹¹¹

Impact of the 9th Edition TNM Classification

New categories: The 9th edition International Association for the Study of Lung Cancer (IASLC) Tumor, Node, Metastasis (TNM) system distinguishes between single-station (N2a) and multistation (N2b) involvement, reinforcing the importance of comprehensive mediastinal staging.^{112,113}

Impact on sampling approach: Under older guidelines, operators may have concluded CP-EBUS staging after confirming one positive N2 station by ROSE. Changes with the 9th edition emphasize the need to sample additional N2 nodes regardless of the initial ROSE result to distinguish between single-station and multistation disease.¹¹³

Practical changes:

ROSE and fresh needles may be increasingly used to reduce cross-contamination once a malignant node is identified. Future protocols may require systematically sampling all N2 stations rather than stopping at the first positive station to ensure accurate staging, prognosis, and treatment decisions.

Recommendation 6: Evidence-Based (Meta-Analysis)

PICO Question

Should mediastinoscopy be performed after a negative CP-EBUS in patients with risk factors for occult mediastinal disease but no overt radiographic involvement?

Recommendation

Do not routinely offer preoperative surgical mediastinoscopy to patients with a negative CP-EBUS-guided mediastinal staging examination who have risk factors for occult N2/N3 disease but no overt radiographic involvement.

Background

Accurate mediastinal staging is crucial for guiding NSCLC treatment. While cervical mediastinoscopy was once the gold standard for preoperative invasive staging, CP-EBUS has largely replaced it, demonstrating comparable accuracy with lower morbidity.^{110,111,114} The negative predictive value of CP-EBUS varies across studies and settings. Very high-probability lymph nodes (large and FDG-avid) with a negative CP-EBUS result (no cancer or an alternative diagnosis like granuloma) should be followed by mediastinoscopy for confirmation. However, there is less clarity regarding confirmatory mediastinoscopy in patients at high-risk of occult disease (eg, cN1 nodes, central tumors, or lesions larger than 3 cm). Balancing the small false-negative risk of CP-EBUS against the costs, risks, and potential surgical delays associated with mediastinoscopy is key.

Summary of Evidence

Three studies ($n = 562$) were identified for quantitative synthesis (Fig. S6, Supplemental Digital Content 1, <http://links.lww.com/LBR/A364>)—1 RCT³⁸ and 2 high-quality prospective observational study (Table 8).^{39,40} The studies evaluated immediate surgical resection versus confirmatory mediastinoscopy after negative CP-EBUS staging. CP-EBUS generally demonstrated relatively high negative predictive values exceeding 90%. In the largest study by Bousema et al,³⁸ unforeseen N2 disease was discovered in 8.8% of patients undergoing immediate resection compared with 7.7% in those undergoing mediastinoscopy, indicating CP-EBUS alone was non-inferior to combined CP-EBUS and mediastinoscopy. Similarly, Lucena et al³⁹ (2023) noted only a 2% additional detection rate with mediastinoscopy in a population with moderate risk for occult metastasis. Finally, Yasufuku et al⁴⁰ (2011) observed similar sensitivity (81% vs. 79%) and negative predictive value (91% vs. 90%) when comparing CP-EBUS alone to CP-EBUS plus mediastinoscopy. With pooled analysis, there was no significant difference in missed N2/N3 disease between the 2 approaches [RD: 0.03 (−0.03 to 0.09), $P = 0.18$, Fig. 5]. The number needed to treat (NNT) ranged from 24 to 77, underscoring limited clinical benefit from adding mediastinoscopy in this patient population.

TABLE 9. Selected Studies Examining Tumor Size (> 3 cm) and Occult Mediastinal (N2) Metastasis

Study	Study design	Population	Tumor				OR (95% CI)	Key findings
			> 3 cm (n)	> 3 cm (n)	≤ 3 cm (n)	≤ 3 cm +N2 (%)		
Al-Sarraf et al (2008) ⁴¹	Retrospective	NSCLC w/ negative mediastinum on PET-CT	87	19.5%	66	12.1%	1.76 [0.71-4.37]	Sixteen percent total occult N2; tumor > 3 cm correlated with LN risk.
Cho et al (2018) ⁴²	Retrospective	NSCLC presenting as subsolid tumor	43	7.0%	203	1.0%	7.54 [1.22-46.57]	Larger solid portion predicted nodal involvement.
Farjah et al (2013) ⁴³	Retrospective	T1/T2N0M0 NSCLC by PET	176	13.6%	449	8.9%	1.61 [0.94-2.77]	Risk rose with tumor size.
Gao et al (2017) ⁴⁴	Retrospective	cT1-2N0 NSCLC by PET-CT	82	8.5%	141	2.1%	4.29 [1.08-17.09]	T2 tumors more likely to have occult N2.

cT1-2N0M0 indicates clinical T1-2 node-negative M0 disease; LN, lymph node; N2, mediastinal metastasis; NSCLC, nonsmall cell lung cancer; OR, odds ratio; PET-CT, positron emission tomography-computed tomography.

Rationale

Although mediastinoscopy may slightly increase the detection of occult mediastinal disease, its additional yield remains small. Given procedural risks, costs, and potential surgical delays, most patients with negative CP-EBUS and no overt radiographic mediastinal disease will not benefit from routine preoperative mediastinoscopy. Instead, surgical resection with intraoperative lymph node dissection is generally preferred. Confirmatory mediastinoscopy may be selectively performed for patients with particularly high-risk imaging features (eg, intense FDG-avid nodes) or incomplete CP-EBUS examination.

Implementation Considerations

- Patient selection: Reserve mediastinoscopy for cases with discordant CP-EBUS-to-imaging results (eg, nondiagnostic CP-EBUS of large or FDG-avid lymph nodes in a high-risk patient) or suspicious findings not thoroughly sampled by CP-EBUS.
- CP-EBUS protocol: A systematic CP-EBUS-guided approach (sampling multiple nodal stations) should be performed to minimize false negatives; see Recommendation 5 for details.
- Resource utilization: In centers with high CP-EBUS expertise, routine mediastinoscopy offers minimal added value and can be safely omitted.
- Postoperative management: If unexpected N2/N3 disease is discovered after resection, established adjuvant therapy protocols apply.

Recommendation 7: Evidence-Based (Meta-Analysis)

PICO Question

Does tumor size using a 3 cm cutoff affect the likelihood of occult mediastinal metastasis in patients with peripheral tumors and a radiographically normal hilum and mediastinum?

Recommendation

Clinicians should recognize a primary tumor size > 3 cm as a significant risk factor for occult mediastinal metastasis and consider invasive nodal staging in patients with peripheral tumors and a radiographically normal hilum and mediastinum.

Note: This recommendation includes risk-stratification guidance rather than a direct “use/do not use” statement. The following uses NICE’s “Consider” to convey that invasive nodal staging should be individualized based on tumor size and clinical context.

Background

While there is general agreement among societies and previous guidelines regarding the indications for invasive lung cancer staging, contention exists about how primary tumor size impacts risk for occult mediastinal involvement. According to the 2013 ACCP guidelines, tumor size is not an independent criterion for invasive staging, while the 2014 European Society of Thoracic Surgeons (ESTS) guidelines specifically include tumor size > 3 cm as an indication.^{110,111}

Summary of Evidence

A meta-analysis of 4 retrospective cohort studies (n = 1247; Table 9) found a significantly higher risk of occult

TABLE 10. Key Studies Examining LN Metastasis in Ground-Glass and Subsolid Nodules

Study	Design / Sample Size	Population / Inclusion	Nodule Characteristics	LN Metastasis Rate	Key Findings
Hayashi 2017 ¹²⁰	Retrospective; n = 188	Pathologic T1 adenocarcinoma ≤ 3 cm	Solid component on mediastinal-window CT ≤ 5 mm vs. > 5 mm	0% when solid ≤ 5 mm	Nearly pure-GGO lesions show no nodal spread; solid thickness is the critical predictor.
Lin 2020 ⁴⁶	Retrospective; n = 768	Stage I lung adenocarcinoma with any GGO	Pure-GGO, GGO-predominant (CTR ≤ 0.5), Solid-predominant (CTR > 0.5)	0% / 3.8% / 6.9%	Pure-GGO group had zero LN spread; risk rises with increasing solid ratio; lymphadenectomy may be unnecessary for pure-GGO.
Moon 2016 ⁴⁷	Retrospective; n = 358	cT1N0M0 NSCLC ≤ 3 cm	GGO-predominant vs. Solid-predominant	0.8% vs. 10.9%	LN-positive disease restricted to solid-predominant tumors; mediastinal LN evaluation provides no benefit in GGO-predominant cases.
Sim 2014 ⁴⁸	Retrospective; n = 48	Surgically resected pure GGO nodules (median 12 mm)	Pure-GGO only	0%	Systemic node sampling revealed no nodal involvement in any pure-GGO patient.
Suzuki 2018 ⁴⁹	Retrospective; n = 160	GGO-dominant (> 3 cm total size)	CTR ≤ 0.25 (Type A), 0.25-0.5 (Type B), > 0.5 (Type C)	0% (Types A & B); 32% in Type C	Even large (> 3 cm) GGO-dominant lesions remain node-negative until the solid portion exceeds 50%.
Wang 2020 ⁵⁰	Retrospective; n = 327	Stage IA peripheral adenocarcinoma presenting as GGO	Pure-GGO, GGO-predominant, Solid-dominant, Pure-solid	0% / 0% / 7.5% / 14.9%	All LN-positive cases occurred in lesions with substantial solid components.
Xu ⁵¹ 2023	Two-center retrospective cohort n = 864 + pooled review (35 studies)	NSCLC ≤ 3 cm (pure & semisolid GGOs)	CTR ≤ 0.5 vs. > 0.5	0–0.1% vs 30.4% (pooled); 0% vs 6.6% (own cohort)	Large, pooled data set confirms near-zero LN risk below CTR 0.5 and sharp rise above that threshold.
Ye 2019 ⁵²	Retrospective; n = 911	Stage I adenocarcinoma (part-solid, pure-GGO, pure-solid)	Part-solid, Pure-GGO, Pure-solid	2.2% / 0% / 27%	Part-solid nodules carry modest LN risk; pure-GGO again zero.
Zhang 2020 ⁵³	Retrospective; n = 2749	Invasive NSCLC up to 5 cm	CTR ≤ 0.5 vs. > 0.5	0% vs. 24.1%	Large series demonstrating CTR ≤ 0.5 as a reliable surrogate for node-negative status.

cT1N0M0 indicates clinical T1 N0 M0 stage (TNM classification); CT, computed tomography; CTR, consolidation-to-tumor ratio (solid-portion diameter $\div$ overall lesion diameter); GGO, ground-glass opacity; LN, lymph node; NSCLC, nonsmall cell lung cancer.

N2 disease in peripheral NSCLC tumors > 3 cm compared with those ≤ 3 cm [RD: 0.06 (0.04-0.07), $P=0.0016$, Fig. 6]. Across individual studies, unsuspected N2 prevalence ranged from ~3% to 19%. Other reports suggest alternate size cutoffs (eg, 2.5 cm) or note additional risk factors, including subsolid lesions and adenocarcinoma subtype.^{7,115-117} Relevant literature review and quality evaluations are summarized in Supplementary Figure S7, Supplemental Digital Content 1, <http://links.lww.com/LBR/A364>.

Rationale

Larger tumors generally demonstrate more aggressive biological behavior and increased lymphatic spread.⁴³ Conventional imaging often fails to detect micrometastatic disease in tumors > 3 cm.⁴⁴ Adopting the 3 cm threshold aligns with TNM staging and helps ensure that higher-risk patients receive thorough nodal assessment. Subgroup analyses suggest features like central location, lower-lobe position, or large solid components further elevate risk.⁴²

Implementation Considerations

- Patient selection: Tumors larger than 3 cm should be considered an independent criterion for invasive mediastinal staging, even in the absence of other known risk factors for mediastinal metastasis.
- Additional risk factors: Other risk factors, such as suspicious N1 uptake and central tumor location, further heighten concern for nodal involvement.
- Future directions: Emerging AI and radiomics approaches may refine thresholds for invasive staging and risk stratification.

Recommendation 8: Evidence-Informed (Systematic Review)

PICO Question

Should invasive mediastinal staging be performed for patients with pure ground-glass nodules (GGNs) and a radiographically normal mediastinum?

Recommendation

Do not routinely offer invasive mediastinal staging for patients with pure GGNs and a radiographically normal mediastinum by CT and PET scan.

Background

GGNs are increasingly detected through incidental findings and screening programs.¹¹⁸ Most pure GGNs represent preinvasive or minimally invasive adenocarcinomas with an indolent course and favorable prognosis.¹¹⁹ Current guidelines focus on tumor size, location, and suspicion of N1 disease to assess a patient's need for mediastinal staging, but offer limited attention to nodule density.^{110,111} Aggressive staging practices for low-risk lesions such as GGNs may lead to unnecessary invasive procedures.

Summary of Evidence

A systematic review was conducted (Supplementary Fig. S8, Supplemental Digital Content 1, <http://links.lww.com/LBR/A364>). Three studies were included in the qualitative synthesis, with only one providing quantitative data. A broader review of 9 retrospective studies ($n=48$ to 2749) consistently demonstrated negligible or absent

mediastinal lymph node (LN) metastases (0% to 1%) from pure GGNs.^{46-53,120} In contrast, part-solid nodules showed a slightly higher—though still low—metastatic LN incidence (~1.6% to 8%), particularly in nodules with larger solid components [consolidation-to-tumor ratio (CTR) > 0.5]. Table 10 summarizes these findings, illustrating near-zero metastasis risk for pure GGNs and highlighting the low, yet higher, risk in part-solid nodules.

Rationale

Pure GGNs typically represent preinvasive or minimally invasive adenocarcinomas with a very low chance of lymphatic spread, rendering additional invasive staging (CP-EBUS, mediastinoscopy) unlikely to alter management. Part-solid nodules with CTR ≤ 0.5 similarly show very low LN metastatic potential. However, part-solid nodules with CTR > 0.5 demonstrate substantially greater risk and generally warrant standard staging approaches.

Implementation Considerations

Accurate radiologic assessment: Use high-resolution thin-section CT to precisely measure the nodule's total diameter and the solid component, thus determining the CTR.

Risk stratification: Invasive staging can likely be omitted for pure ground-glass nodules or those with a small solid component (CTR ≤ 0.5). Conversely, if CTR is > 0.5 or CT/PET suggests nodal disease, standard mediastinal staging (CP-EBUS, mediastinoscopy) should be performed.

Multidisciplinary coordination: Ensure standardized terminology and measuring techniques are used by radiologists, thoracic surgeons, and oncologists regarding CT.

Future Directions: AI and radiomics may improve risk stratification and refine invasive staging thresholds.

Recommendation 9: Evidence-Informed (Systematic Review)

PICO Question

Should CP-EBUS be performed to detect occult N1 disease before stereotactic body radiation therapy (SBRT) in patients with early-stage NSCLC?

Recommendation

We make no recommendation regarding the routine use of CP-EBUS to detect occult N1 disease before SBRT in patients with early-stage NSCLC who do not meet criteria for invasive staging.

Background

SBRT is a favored treatment option for many patients with early-stage NSCLC, particularly those who have significant comorbidities or may not be suitable surgical candidates. While outcomes for well-selected SBRT candidates are typically excellent, roughly 10% to 20% will develop locoregional recurrence.^{121,122} A primary difference between these patients and those undergoing surgery is that the latter benefit from pathologic nodal assessment, whereas SBRT patients generally do not receive invasive staging. Although risk factors for occult N2/N3 disease are well documented, the same is not true for occult N1.¹²³ Undetected N1 disease, which would remain untreated with SBRT, may be a significant driver of recurrence.^{60,124}

TABLE 11. Key Retrospective Studies Evaluating EBUS Before SBRT for Early-Stage NSCLC

Study	Design/sample size	Population/key inclusion criteria	Study focus and methods	Comparator groups	Diagnostic yield/treatment efficacy	Key outcomes	Summary/conclusions
Akthar et al (2018) ⁵⁴	Retrospective; N = 92	cStage I (T1–T2, N0, M0) NSCLC; all received SBRT; some underwent CP-EBUS	Compared CP-EBUS-staged vs. no CP-EBUS-staged for regional control, DFS, and OS	CP-EBUS (n = 37) vs. No CP-EBUS (n = 55)	Up-staging not reported; 2 y FFRF 86% vs. 88%, OS 88% vs. 77%	No difference in regional failure, DFS, or OS	Routine EBUS offered no outcome benefit when imaging was cN0; inability to assess occult N+ because EBUS group was node-negative by design
George et al (2024) ^{55*}	Retrospective; N = 376	Stage IA–II (T1–T2, N0) by imaging; all received SBRT; median follow-up ~43 mo	Examined the correlation of CP-EBUS-based nodal staging with OS, recurrences	INS (CP-EBUS only) (n = 142) vs. No INS (n = 234)	No recurrence benefit; multivariable OS HR: 1.66 (1.17–2.35)	INS associated with worse OS, likely selection bias	Universal EBUS not supported; advocate selective use
Harris et al (2018) ⁴⁵	NCDB-based; N = 24,603	cT1–T2, N0, M0 NSCLC, 7% had INS (CP-EBUS ± med), all treated with SABR (2004–2014)	Used propensity/instrumental variable analyses to assess the OS impact of INS	INS vs. No INS	Median OS 2.8 y vs 2.7 y; IV-adjusted HR: 1.0	No survival benefit	Large cohort supports selective rather than routine INS
Kennedy et al (2020) ⁵⁶	Retrospective; N = 651	cT1–T2, N0, M0 NSCLC (PET-CT negative nodal disease), all received SBRT	Assessed 2 y nodal/distant control, OS with vs. without invasive staging	INS (CP-EBUS or med) vs. No INS	Nodal + distant control 75% vs. 78%; OS identical	INS conferred no control or OS gain	Routine INS not useful when PET-CT is negative
Mullins et al (2021) ⁵⁷	Retrospective; N = 158	cT1–T2, N0, M0 NSCLC by imaging; all treated with SBRT	Compared event-free/OS outcomes for CP-EBUS vs. imaging staging	CP-EBUS vs. No CP-EBUS	Three year nodal-failure freedom 84% vs. 69% (<i>P</i> = 0.10); OS 65% vs. 67%	Nonsignificant trend to better nodal control; no OS gain	Authors favour selective staging only
Ong et al (2015) ⁵⁸	Retrospective; N = 220	cT1–2a, N0, M0 (< 5 cm) by integrated PET-CT; all underwent CP-EBUS	Evaluated CP-EBUS sensitivity vs. imaging; no separate “no CP-EBUS” group	Single cohort (CP-EBUS for all)	Sens 36.7%, Spec 100%, NPV 84.7%; 22% overall occult N+ (8% detected by EBUS)	EBUS detects some occult nodes but misses many in inaccessible stations	Useful adjunct before ablative therapy, yet limited by reach
Schonewolf et al (2018) ⁵⁹	Retrospective; N = 236	cT1–T2, N0, M0 NSCLC; ~24% had IMNS (CP-EBUS or med) pre-SBRT	Compared local, nodal, distant failures & OS for IMNS vs. no IMNS	IMNS vs. No IMNS	Isolated nodal failure 14% vs. 8% (NS); OS HR 0.70 (0.43–1.12)	IMNS did not improve recurrence or OS	Restrict invasive staging to selected high-risk cases
Vial et al (2017) ⁶⁰	Retrospective; N = 246	cT1–2a, N0, M0 (< 5 cm) by PET-CT; all potential SABR candidates	Compared PET-CT vs. CP-EBUS for nodal (up/down) staging; single-cohort	Single cohort (CP-EBUS for all)	19% stage shift; in PET-N0 pts 3.4% occult N+ found by EBUS	Stage migration common, yet OS/PFS unchanged	EBUS refines stage but outcome impact unproven

*Study published after the March 2023 literature search cutoff; included by post-hoc panel consensus due to high relevance and impact.

cN0 indicates clinically node-negative on imaging; CP-EBUS, convex-probe endobronchial ultrasound; DFS, disease-free survival; DMFS, distant metastasis-free survival; FFRF, freedom from regional failure; HR, hazard ratio; IMNS, invasive mediastinal nodal staging; INS, invasive nodal staging; LRFS, local recurrence-free survival; Med, mediastinoscopy; NCDB, National Cancer Database; NPV, negative predictive value; NRFS, nodal recurrence-free survival; NSCLC, nonsmall cell lung cancer; OS, overall survival; PET-CT, positron emission tomography–computed tomography; PFS, progression-free survival; RFS, recurrence-free survival; SABR, stereotactic ablative radiotherapy; SBRT, stereotactic body radiation therapy.

On this basis, routine CP-EBUS assessment—even for patients with small, peripheral tumors and seemingly minimal risk of occult N2 disease—has been adopted at many institutions as a means of detecting N1 disease and potentially reducing the risk of SBRT failure. However, there is no conclusive evidence that identifying N1 disease by CP-EBUS staging improves overall survival or reduces recurrence. Moreover, CP-EBUS adds cost, procedural risks, and the possibility of delaying definitive therapy.

Summary of Evidence

No studies met criteria for a quantitative synthesis (Supplementary Fig. S9, Supplemental Digital Content 1, <http://links.lww.com/LBR/A364>). A review of 8 retrospective evaluations (Table 11) involving patients with clinically node-negative (cN0) NSCLC who received SBRT with or without invasive nodal staging (INS) was performed. Among studies that specifically evaluated outcomes for INS versus no INS in cN0 patients (as determined by PET/CT), there was no consistent evidence of a survival advantage or a decrease in recurrence attributable to INS.^{45,54,55,57,59} In most of these cohorts, patients were either deemed N0 through CP-EBUS or received no staging at all, so the actual detection rates of occult N1 were not extensively documented.

In 2 studies, roughly 2% to 5% of cN0 patients were upstaged by CP-EBUS to N1.^{58,60} In the series by Ong and colleagues, which included surgical patients with cN0 disease, CP-EBUS uncovered N1 disease in 5%, whereas intraoperative assessments eventually revealed N1 involvement in ~20% of patients. Notably, 70% of these “missed” nodal stations were intralobar (stations 12, 13, 14) and therefore beyond CP-EBUS reach.

Rationale

Although CP-EBUS can detect occult N1 disease in a subset of clinically node-negative (cN0) patients, the available evidence is currently limited and does not conclusively show whether doing so solely to rule out N1 disease before SBRT improves survival or reduces recurrence. Many patients undergoing SBRT have significant comorbidities or cannot tolerate surgery; thus, identifying isolated N1 disease may not substantially alter management or impact outcomes in this setting. Any potential benefit must also be weighed against the time, cost, and procedural risks of invasive staging in a population often at higher procedural risk.

Importantly, this review does not address patients whose imaging is suspicious for N2/N3 disease or who meet standard invasive staging criteria. In addition, patients who opt for SBRT over surgery despite having fewer comorbidities and a longer expected survival may represent a distinct population where the impact of occult N1 disease could differ; however, these groups were beyond the scope of the current analysis. Due to the limited number and quality of studies, we cannot make a definitive recommendation for or against routine CP-EBUS staging in patients who appear to have early-stage disease on imaging.

Implementation Considerations

- **Limited impact on management:** While identifying occult N1 disease may alter treatment strategies in certain cases, current data do not conclusively show that routine CP-EBUS staging improves survival or disease control for early-stage NSCLC patients treated with SBRT.

- **Risk/benefit assessment:** Any benefit gained from detecting occult N1 disease must be balanced against the time, cost, and procedural risks associated with CP-EBUS—especially in patients with multiple comorbidities who are already considered poor surgical candidates.
- **Select populations:** For individuals with established risk factors for N2/N3 disease (eg, suspicious lymph node features on imaging, central tumor location, tumor size > 3 cm), standard invasive staging is still recommended before SBRT, in accordance with existing guidelines.

Future Research: Further data are needed to define CP-EBUS’s full utility in this patient population, particularly as newer, smaller-footprint CP-EBUS scopes become available. Future investigations should examine whether more comprehensive sampling of intralobar nodes (eg, stations 12 to 14) can improve detection rates and ultimately translate to better outcomes for patients with early-stage NSCLC.

CONCLUSIONS

These evidence-based and evidence-informed guidelines represent the first comprehensive AABIP recommendations focusing specifically on the bronchoscopic diagnosis and staging of lung cancer. Topics for evaluation were chosen based on clinical and potentially actionable relevance and target those who are most likely to perform invasive tissue evaluation of lung cancer. The goal was to provide a framework based on the best available evidence and, where data were lacking, expert consensus. To ensure an unbiased, scientifically valid, standards-compliant, and trustworthy resource, a methodologically rigorous process combining GRADE evidence appraisal with NICE-structured recommendations was utilized.

We acknowledge several limitations to our recommendations. Much of the supporting evidence for the recommendations stems from retrospective analyses and observational studies, with relatively few high-quality randomized controlled trials. This limited the strength of several recommendations. Therefore, it is important to emphasize that a lack of strict adherence to these guidelines should not be viewed as a deviation from the standard of care.

A significant limitation of our meta-analysis is the variable methodology used to define and calculate diagnostic yield and accuracy across studies. As noted by Gonzalez et al,⁷³ the reported diagnostic yield can vary by more than 20% depending on whether strict, intermediate, or liberal definitions are used. This methodological heterogeneity limits the interpretability of our pooled estimates and highlights the importance of adherence to standardized outcome reporting in future bronchoscopy studies.

The rapid evolution of PNB guidance technology concurrent with and subsequent to our literature search parameters resulted in various methodologic constraints and may limit the longevity of some conclusions. We addressed these by updating the literature review and analysis as new data became available, incorporating indirect evidence when applicable, and discussing relevant knowledge and evidence gaps. Also, a substantial proportion of our guidelines addresses CP-EBUS practice, which has an established evidence base that was not subject to as many rapid changes

as PNB. Furthermore, many of our recommendations address sampling concepts that are mostly modality-neutral for PNB guidance, preserving the relative applicability and relevance of the guideline. Nevertheless, future iterations will likely require recalibration as more robust performance data on these and other emerging technologies relevant to advanced diagnostic bronchoscopy and PPN evaluation become available.

Since the completion of our search (March 2023), several high-impact studies have evaluated robotic-assisted bronchoscopy, real-time cone-beam CT, and digital tomography-synthesis. Preliminary results suggest diagnostic yields approaching those of CT-guided biopsy with lower complication rates. Although promising, these data remain insufficiently mature for an evidence-graded recommendation. We acknowledge this evidence gap as a significant limitation of the current guidelines, and the panel has prioritized a focused future update to address this question.

Our guideline working and expert group panel lacked a medical oncologist, pathologist, and patient advocate. We recognize that diversifying representation is a fundamental aspect of guideline development and that these stakeholders would contribute valuable perspectives to lung cancer diagnosis and management in general. However, this effort uniquely focused on specific technical aspects of and approaches to diagnostic procedures for lung nodules and cancer, which were most relevant to those performing them. A broader inclusion of lung cancer stakeholders who would provide insight into therapeutic implications, long-term clinical trajectories, and quality-of-life endpoints is encouraged for guidelines of a broader scope.

Our process identified multiple areas in need of more rigorous investigation. While the randomized VERITAS trial⁷⁴ represents a major advancement in providing high-quality data on the diagnostic performance of minimally invasive techniques, comparing newer PNB technologies with percutaneous sampling remains a high priority. Standardization of tissue acquisition protocols for optimal molecular testing likewise warrants further exploration. Prospective evaluation of optimal staging thresholds for patients with a radiographically normal mediastinum, as well as the utility of systematic versus targeted sampling strategies, is another critical domain for future research.

The AABIP intends this guideline to function as a living document. A formal evidence update is planned within 3 to 5 years, with consideration for an earlier targeted revision should a sufficient quantity of high-quality comparative data on robotic-assisted bronchoscopy or real-time intraprocedural imaging become available.

In conclusion, these guidelines were developed to provide practical and useful assistance to clinicians performing diagnosis and staging in patients with suspected or confirmed lung cancer. We encourage adaptation of these recommendations in accordance with local expertise, institutional capabilities, and patient-specific factors. Recognizing that this field is rapidly evolving, the periodic reassessment and updating of these guidelines will be needed. We hope this document provides a valuable summary of existing evidence and expert-based perspectives, stimulates future investigation, and facilitates more robust, evidence-based guidance.

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